

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
LOS ANGELES REGION**

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**ORDER NO. R4-2020-0049  
FILE NO. 14-031  
CI-10041**

**WASTE DISCHARGE REQUIREMENTS AND  
WATER RECLAMATION REQUIREMENTS  
ISSUED TO  
CITY OF SANTA CLARITA  
(VISTA CANYON WATER FACTORY)**

The California Regional Water Quality Control Board, Los Angeles Region (Regional Water Board) finds:

**BACKGROUND**

1. The Vista Canyon Project proposes to develop an approximately 185-acre area, providing 1,100 residential units with a residential population estimated at 3,500, as well as up to 950,000 square feet of commercial and medical offices, retail stores, a theater, restaurants, and a hotel. Approximately 392,000 gallons per day (GPD) of wastewater will be generated from this project once it is fully developed. Full development is projected to be completed between 2025 and 2027 based upon market conditions.
2. The Vista Canyon Water Factory (Water Factory) is a tertiary wastewater treatment and recycling plant that will treat wastewater generated from the Vista Canyon Project. This treated wastewater will be recycled for onsite and offsite landscape irrigation and other non-potable applications. When use of recycled water for irrigation during rainy weather is prohibited or when the effluent cannot meet effluent limitations prescribed in this Order, effluent will be conveyed to the Saugus Water Reclamation Plant (WRP) and the Valencia WRP, owned and operated by Santa Clarita Valley Sanitation District (SCVSD). The Saugus WRP will be the primary plant treating wastewater. The Valencia WRP will be the backup plant to treat any extra wastewater generated beyond the wastewater treatment capacity of the Saugus WRP.
3. The Water Factory will be operated by the City of Santa Clarita (City). The City, hereafter defined as Permittee or Discharger, will be the owner of the Water Factory

and will be responsible for the treatment of wastewater, wastewater quality, recycled water quality, and any groundwater quality impacted by the discharge and the recycled water applications. The City will also be responsible for compiling and submitting all monitoring data and reports to the Regional Water Board.

4. On July 9, 2019, the City adopted the City of Santa Clarita Recycled Water Ordinance to provide for the establishment and enforcement of regulations on the proper uses, sale and distribution of recycled water produced at the Water Factory. The adopted ordinance prohibits the sale, distribution, or use without a Wholesale-Recycled Water Purchase Agreement (Agreement). The ordinance further expressed the City's intent to enter into an Agreement with the Santa Clarita Valley Water Agency (SCVWA) for the distribution of recycled water produced at the Water Factory.
5. The Regional Water Board conducted a Vista Canyon Project site visit on August 9, 2019. The site is along the Santa Clara River with an estimated distance of 500 feet southeast to the center of the Santa Clara River. The Water Factory construction was completed in May 2019. During the visit, Board staff observed the Water Factory in temporary operation using potable water that is recirculated.

## **REGULATORY AGENCIES**

6. The Regional Water Board is the permitting agency for the discharge of disinfected, tertiary-treated effluent from the Water Factory via non-potable recycled water applications. This Regional Water Board issues Waste Discharge Requirements (WDRs) and Water Reclamation Requirements (WRRs) to ensure that discharges of recycled water from the Water Factory do not adversely affect the quality of groundwater and its beneficial uses.
7. The Regional Water Board is required pursuant to California Water Code (CWC) section 13523 to consult with and receive recommendations from the Division of Drinking Water (DDW) within the State Water Resources Control Board (State Water Board) (formerly within the California Department of Public Health) regarding public health, safety, or welfare.

## **PURPOSE OF ORDER**

8. On June 9, 2016, the Regional Water Board adopted WDRs and WRRs, as Order No. R4-2016-0220, to regulate the proposed discharge of disinfected, tertiary-treated wastewater generated at the Water Factory for non-potable recycled water applications.
9. On March 14, 2019, the City requested an amendment of the WDRs and WRRs to reevaluate limits established for the Water Factory based on recent potable water supply and ambient groundwater quality data. The City subsequently submitted a

Report of Waste Discharge dated June 4, 2019. A review of the recent potable water supply and groundwater quality data indicated that the previous effluent and groundwater limitations should be updated.

10. The purpose of this Order is to renew WDRs and WRRs for the City's Water Factory. This Order includes updates of the wastewater treatment process, effluent and groundwater limitations based on updated water supply and ambient water quality data, and groundwater impact analysis to ensure that the City's discharge of waste complies with water quality objectives and requirements set forth in *Water Quality Control Plan for the Los Angeles Region: Basin Plan for the Coastal Watersheds of Los Angeles and Ventura Counties* (Basin Plan) for the protection of beneficial uses.
11. CWC section 13260 requires any person "proposing to discharge waste, within any region that could affect the quality of the waters of the state, other than to a community sewer system," to file a report of waste discharge. The term "waste" is defined in CWC section 13050(d) to include "sewage and any and all other waste substances, liquid, solid, gaseous, or radioactive, associated with human habitation, or of human or animal origin, . . . prior to, and for purposes of, disposal." The Discharger proposes to discharge tertiary-treated human sewage, i.e., "waste" to land where it could affect the quality of the waters of the state. Sewage contains various waste constituents, including total dissolved solids (TDS), sulfate, salts (e.g., chloride, boron), bacteria, nitrogen, priority pollutants, and constituents of emerging concern. In accordance with CWC section 13263(g), no discharge of waste into waters of the state, whether or not the discharge is made pursuant to waste discharge requirements, shall create a vested right to continue the discharge. All discharges of waste into waters of the state are privileges, not rights.
12. CWC section 13263 authorizes the Regional Water Board, after any necessary hearing, to prescribe requirements as to the nature of any proposed discharge with relation to the conditions existing in the disposal area or receiving waters upon, or into which, the discharge is made or proposed. The requirements must implement any relevant water quality control plans that have been adopted, and shall take into consideration the beneficial uses to be protected, the water quality objectives reasonably required for that purpose, other waste discharges, the need to prevent nuisance, and the provisions of CWC section 13241.
13. CWC section 13267 authorizes the Regional Water Board to require that any person who proposes to discharge waste to furnish, under penalty of perjury, technical or monitoring program reports which the regional water board requires. The burden, including costs, of these reports, bears a reasonable relationship to the need for the report and the benefits to be obtained from the reports. This Order incorporates Monitoring and Reporting Program (MRP) No. CI-10041 for the City (File No. 14-031), which is necessary to ensure that the discharge of waste, and the use of

recycled water, complies with this Order and is protective of human health and the environment.

14. This Order is adopted pursuant to CWC sections 13263, 13267, and 13523. It sets forth requirements, prohibitions, and other conditions to implement the Basin Plan; prescribes the limits for the recycled water and the Discharger's responsibilities for the production, distribution, monitoring, and application of recycled water; and includes an MRP. The Discharger is responsible for inspecting point-of-use facilities and ensuring compliance with the WDRs and WRRs contained in this Order.

## **VISTA CANYON PROJECT**

### **15. Vista Canyon Project Vicinity**

- A. The Vista Canyon Project (Figure 1) is located in the Santa Clarita Valley within the City. The Vista Canyon Project is immediately south of State Route 14 (SR-14), west of La Veda Avenue, north of the Metrolink rail line, and east of the Colony Townhome community.
- B. The Vista Canyon Project overlies the Santa Clara River Valley East Groundwater Basin (DWR Basin No. 4-4.07) (Figure 2) as defined in the Basin Plan, specifically within the Santa Clara-Mint Canyon area. For management purposes, the Santa Clara River Valley East Groundwater Basin is divided into six subunits/management zones, which exhibit consistent hydrological, water quality or overlying land use characteristics. The Santa Clara-Mint Canyon area is identified as Management Zone 1.

### **16. Water Factory**

- A. The Water Factory (34° 24' 51.73" N, 118° 26' 22.58" W) is located in the southwest corner of the Vista Canyon Project and approximately 200 feet from Santa Clara River (Figure 1). The Water Factory construction was completed in May 2019.
- B. Wastewater generated from the Vista Canyon Project is conveyed by gravity flow to the Water Factory.
- C. The Water Factory has an average design flow capacity of 392,000 GPD, which is capable of generating 371,000 GPD of effluent to be recycled. The wastewater treatment process consists of primary treatment (vertical fine screen), secondary treatment with a retention time of 20 hours (two aeration tanks with nitrification and denitrification activated sludge, reducing ammonia and nitrate concentrations), clarification with a retention time of 0.5 hour (two sedimentation tanks with coagulation and flocculation, reducing solids), tertiary treatment (disc filters, reducing turbidity, and suspended solids)

greater than 85%), and disinfection (UV and chlorination with sodium hypochlorite) (see Figure 3 for process flow schematic). The sludge (21,000 GPD), any excess treated effluent not recycled from the Water Factory, and any untreated wastewater generated from the Vista Canyon Project is discharged to the Saugus WRP and the Valencia WRP. Any effluent not meeting effluent limitations is delivered to the headworks for further treatment.

D. Treated Effluent/Recycled Water Applications

- a. The treated effluent/recycled water will be stored in two 100,000-gallon recycled water storage tanks (Figure 4). Recycled water will be distributed via the recycled water pump station to onsite uses of the Vista Canyon Project for landscape irrigation and other Title 22 non-potable recycled water applications approved by the DDW. This WDR/WRR authorizes treated effluent non-potable recycled water applications in areas overlying the Management Zone 1a of the Santa Clara-Mint Canyon groundwater subbasin. Figure 5 presents onsite recycled water irrigation locations.

A summary of recycled water demand is identified in the Water Factory's Title 22 Engineering Report dated November 27, 2019 as shown in Table 1. SCVWA will be the responsible agency for the conveyance, storage, and onsite and offsite use of the recycled water diverted to its facilities. A separate Title 22 Engineering Report and separate water recycling requirements would be needed for the offsite uses. When the recycled water demand exceeds the supply available from the Water Factory, the potable water will be supplied.

**Table 1. Proposed Annual Average Quantities of Recycled Water Applications**

| <b>Recycled Water Demand</b> | <b>Recycled Water Quantity</b> |
|------------------------------|--------------------------------|
| Onsite irrigation            | 77,582 GPD (86.9 AFY)          |
| Offsite irrigation           | 374,926 GPD (420 AFY)          |

Table 1 notes: The unit of AFY denotes acre feet per year.

- b. An additional 1-million gallon recycled water storage tank will be constructed by SCVWA offsite and to the south of the Vista Canyon Project in order to store the treated wastewater during the following conditions:
  - i. When there is no demand for recycled water; and
  - ii. When the volume exceeds the capacity of the two onsite 100,000-gallon recycled water storage tanks.

- c. Excess water not diverted to SCVWA's 1-million gallon offsite storage tank will be sent downstream to either the Saugus or Valencia WRPs.

E. Receiving Water Monitoring Network

The City installed six (6) monitoring wells (MW-1 through MW-6) in September 2016. The City has been monitoring groundwater for a minimum of three years since the 2016 Order was issued, once in 2016, once in 2017, and once in 2018, prior to discharge of effluent from the Water Factory to understand the groundwater quality in the shallow and deep aquifers. The groundwater quality data were incorporated into the monitoring reports and were submitted to the Regional Water Board for review.

Groundwater monitoring wells, specified in Figure 6, are used to ensure that the treated effluent (recycled water) used for landscape irrigation does not cause the groundwater to exceed receiving water limitations set forth in Section III, Table 11 of this Order.

The groundwater monitoring program network implemented by the City consists of a total of nine (9) wells: five (5) upgradient, two (2) downgradient, and two (2) cross-gradient. More information of these groundwater monitoring wells is available in Table 4, section V.3.A of the accompanying MRP No. CI-10041.

## **GLOBAL WARMING AND CLIMATE CHANGE**

17. In Southern California, the predicted impacts of climate change are numerous. Annual average temperatures are expected to increase, coupled with a higher frequency of extreme heat days. A likely consequence of this warmer climate will be more severe drought periods, leading to an increase in the amount and intensity of fires and a longer fire season. In addition, precipitation patterns are likely to be modified. A decrease in snowfall, combined with warmer temperatures, will induce a decrease in the amount and duration of snowpack, an essential source of freshwater to the region. Although changes to mean precipitation are expected to be small, the increasing occurrence of extreme precipitation events will amplify the risk of flooding.

These impacts may affect water quality in multiple ways, including decreases in stream flow, reductions in, and changes to, aquatic habitats, increases in surface water temperature, increases in pollutant levels, sedimentation, algal growth, and changes in salinity levels and acidification in coastal areas. For permitted facilities such as Publicly Owned Treatment Works (POTWs), specific impacts could include, but are not limited to, an increase in the concentration of pollutants entering the facility, an increase in the temperature of effluents and receiving waters, an increase

in storm water inflow and infiltration, increase in flooding inundation of facilities, sewer overflows, power outages, pump maintenance issues, and onsite or nearby hillside destabilization.

On March 7, 2017, the State Water Board adopted Resolution No. 2017-0012, *Comprehensive Response to Climate Change*, recognizing the challenges posed by climate change, directed state agencies to take climate change into account in their planning decisions, guided by the following principles: Priority should be given to actions that both build climate preparedness and reduce greenhouse gas emissions; where possible, flexible and adaptive approaches should be taken to prepare for uncertain climate impacts; actions should protect the state's most vulnerable populations; and natural infrastructure solutions should be prioritized. On May 10, 2018, the Regional Water Board adopted Resolution No. R18-004, *A Resolution to Prioritize Actions to Adapt to and Mitigate the Impacts of Climate Change on the Los Angeles Region's Water Resources and Associated Beneficial Uses*, which encourages mitigating direct and indirect impacts of climate change on water quality and beneficial uses.

On August 26, 2019, the Discharger submitted a Climate Change Effects Vulnerability Assessment and Management Plan (Climate Change Plan) as required in the prior Order. This Order requires the Discharger to periodically review and submit a revised Climate Change Plan when conditions at the facility change that may impact water quality.

18. The Water Factory is within the 100-year floodplain of the Santa Clara River. Therefore, in response to anticipated climate change effects, the City has considered additional flood control measures and protection of the Vista Canyon Project and the Water Factory Project, including raising the elevation of the Water Factory above the floodplain and designing a concrete retaining wall along the bank of the Santa Clara River. The concrete retaining wall is approximately 18 feet plus a freeboard of approximately 3 feet above the Santa Clara River corridor. This wall is designed in conformance with the County of Los Angeles Capital Flood (Qcap) requirements, which exceed a 1,000-year storm event. Climate change may also increase drought and related impacts such as reduced potable water supply. The Water Factory will produce a new source of recycled water for areas of Los Angeles County and the City to reduce the need for potable water use in the area.
19. Waste Discharge Requirements for this facility contain provisions to require planning and actions to address climate-related impacts that can cause or contribute to violations of permit requirements and/or degradation of waters of the state.

## APPLICABLE PLANS, POLICIES AND REGULATIONS

Due to the unique hydrogeological conditions of the Plant location and its vicinity, this permit is consistent with and implements the Basin Plan, Title 22 California Code of Regulations (CCR), and other essential plans, policies, and regulations to protect the receiving groundwater quality.

20. ***Water Quality Control Plan for the Coastal Watersheds of Los Angeles and Ventura Counties (Basin Plan)*** – The Basin Plan (i) designates beneficial uses for surface and groundwater, (ii) establishes narrative and numeric water quality objectives that must be attained or maintained to protect the designated beneficial uses, and (iii) sets forth implementation programs to protect the beneficial uses of the waters of the state. The Basin Plan also incorporates State Water Board Resolution 68-16 “Statement of Policy with Respect to Maintaining High Quality of Waters in California” (also called the “Antidegradation Policy”). In addition, the Basin Plan incorporates by reference applicable State and Regional Water Board plans and policies and other pertinent water quality policies and regulations. This Order implements the plans, policies and provisions of the Regional Water Board’s Basin Plan.

The Basin Plan (Chapter 3) incorporates Title 22 CCR primary maximum contaminant levels (MCLs) by reference (see Finding No. 21 below) as water quality objectives. This incorporation by reference is prospective including future changes to the incorporated provisions as the changes take effect. The Title 22 CCR primary MCLs are applicable water quality objectives for receiving waters designated as municipal and domestic supply. Also, the Basin Plan specifies that “Ground waters shall not contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.” Therefore, the Title 22 CCR secondary MCLs, which are limits based on aesthetic, organoleptic standards, are applicable water quality objectives for receiving waters designated as municipal and domestic supply. These water quality objectives are implemented in this Order to protect groundwater quality.

In addition, the Basin Plan incorporates State Water Board Resolution No. 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. Beneficial uses applicable to the groundwater are shown in Table 2 as follows:

**Table 2. Basin Plan Beneficial Uses of Groundwater**

| Receiving Water                                                                                          | Beneficial Uses                                                                                                                  |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Santa Clara River Valley<br>East Groundwater Basin-<br>Santa Clara-Mint Canyon<br>(DWR Basin No. 4-4.07) | Existing:<br>Municipal and domestic water supply, industrial service supply, industrial process supply, and agricultural supply. |



The water quality objectives applicable to the Santa Clara River Valley East Groundwater Basin-Santa Clara-Mint Canyon are shown in Table 3 as follows:

**Table 3. Water Quality Objectives for the Santa Clara River Valley East Groundwater Basin-Santa Clara-Mint Canyon**

| <b>TDS<br/>(mg/L)</b> | <b>Chloride<br/>(mg/L)</b> | <b>Sulfate<br/>(mg/L)</b> | <b>Boron<br/>(mg/L)</b> | <b>Nitrate as nitrogen<br/>(mg/L)</b> |
|-----------------------|----------------------------|---------------------------|-------------------------|---------------------------------------|
| 800                   | 150                        | 150                       | 1.0                     | 10                                    |

Table 3 notes: The unit of mg/L denotes milligrams per liter.

- A. **Total Maximum Daily Load (TMDL).** To restore impaired water quality and beneficial uses, in 2003, the Regional Water Board amended the Basin Plan to incorporate a TMDL for Nitrogen Compounds for the Santa Clara River (Figure 7). The TMDL assigns load allocations to groundwater discharges among others. The recycled water percolating from the site will reach underlying groundwater, which is likely connected to the Santa Clara River Reach 7. Therefore, the load allocation assigned to groundwater discharges to “other reaches and tributaries” applies to the Water Factory, which discharges to Reach 7 as defined in the Basin Plan (Table 4).

**Table 4. Santa Clara River Nitrogen Compounds TMDL – Load Allocation for Other Reaches and Tributaries**

| <b>Constituent</b>                | <b>Load Allocation</b> |
|-----------------------------------|------------------------|
| Ammonia-N + Nitrate-N + Nitrite-N | 10 mg/L                |

However, the effluent limit (see section II.2) is based on the more stringent groundwater receiving water limit to ensure the protection of groundwater quality based on the antidegradation analysis.

21. **Title 22 CCR –** Title 22 CCR contains primary and secondary MCLs for inorganic, organic, and radioactive contaminants in drinking water. These MCLs are codified in Title 22 CCR. Title 22 primary MCLs have been incorporated into the Basin Plan as water quality objectives (see Attachments B-1 to B-6). MCLs are used as one of the primary bases for effluent limits for discharges of recycled water in WDRs and WRRs to protect the designated beneficial uses of municipal and domestic supply.
22. **Nonpoint Source Implementation and Enforcement Policy –** State Water Board adopted the *Policy for Implementation and Enforcement of the Nonpoint Source Pollution Control Program* (Nonpoint Source Control Policy) on May 20, 2004. The Nonpoint Source Control Policy requires that all sources of nonpoint source pollution be regulated through WDRs, waivers of WDRs, or discharge prohibitions, or some combination of these regulatory tools.

Recycled water applications can adversely impact water quality and impair beneficial uses by contributing excessive contaminants, such as nutrients and salts. These nonpoint source discharges from the Water Factory are discharges of waste that could affect the quality of waters of the State. This Order is consistent with the Nonpoint Source Control Policy because it contains WDRs. This Order requires the Discharger to attain water quality standards through effluent and receiving water limitations and prescribes effluent and groundwater quality monitoring within the recycled water use area to evaluate compliance with limitations.

23. **Publicly Owned Treatment Works (POTW)** – The term POTW means a treatment works as defined by section 212 of the federal Clean Water Act, which is owned by a State or municipality (as defined by section 502(4) of the Act). This definition includes any devices and systems used in the storage, treatment, recycling and reclamation of municipal sewage or industrial wastes of a liquid nature. It also includes sewers, pipes and other conveyances only if they convey wastewater to a POTW treatment facility. The term also means the municipality as defined in section 502(4) of the Clean Water Act, which has jurisdiction over the indirect discharges to and the discharges from such treatment works. (40 Code of Federal Regulations [CFR] 403.3(q)). The Water Factory meets all above criteria and therefore is considered a POTW.
24. Pursuant to CWC section 13263, the requirements of this Order take into consideration the provisions of CWC section 13241, including the following factors.
- A. Past, present, and probable future beneficial uses of water;

The receiving water for discharges from the Water Factory is the Santa Clara River Valley East Groundwater Basin, Santa Clara-Mint Canyon subunit. The receiving water limitations in this Order are specified to maintain the beneficial uses of this basin: municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR). This Order also specifies effluent limitations protective of the beneficial uses and includes effluent and receiving water monitoring and reporting requirements to verify that discharges will not adversely affect the beneficial uses of groundwater.

- B. Environmental characteristics of the hydrographic unit under consideration, including the quality of the water available thereto;

This Order incorporates the site-specific water quality objectives for groundwater in the Basin Plan considering geology, hydrogeology, and hydrology. Based on recent and historical data, the groundwater basin currently has high quality water, but is experiencing increases in salt and nitrogen loading from natural and anthropogenic sources. The Water Factory will

produce effluent quality that is better than the groundwater quality objectives and will comply with the state's Antidegradation Policy (Resolution No. 68-16). The project will therefore limit further groundwater degradation.

- C. Water quality conditions that could reasonably be achieved through the coordinated control of all factors that affect water quality in the area;

As discussed in Findings 30 and 31, the Water Factory is included in the Salt and Nutrient Management Plan for the entire groundwater basin. In addition, the City performed an impact analysis on potential localized impacts due to the proposed discharge. The analysis demonstrates that, with respect to background TDS, chloride, sulfate, and boron, these constituent concentrations will remain below groundwater quality objectives and will comply with the state's Antidegradation Policy.

- D. Economic considerations;

An upgrade of the Water Factory to treat salts and further treat nitrogen compounds would require installation of a reverse osmosis system, which is not currently economically feasible and is unnecessary because the discharge from the Water Factory will not cause adverse impacts to the groundwater quality.

- E. The need for developing housing within the region;

The Water Factory will serve the Vista Canyon Project, which will provide 1,100 residential units with a residential population estimated at 3,500, as well as up to 950,000 square feet of commercial and medical offices, retail stores, a theater, restaurants, and a hotel. This will help address the longstanding housing shortage in California, which ranks among the worst of the 50 states for housing units per capita.

- F. The need to develop and use recycled water;

This Order authorizes the City to treat and discharge up to 371,000 GPD of disinfected tertiary-treated wastewater to generate recycled water for onsite irrigation pursuant to CCR Title 22 section 60304(a). This will help offset the need for potable water.

25. **AB 685 – CWC Section 106.3** – It is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes. This Order promotes that policy by requiring discharges to meet maximum contaminant levels developed to protect human health and ensure that water is safe for domestic use.

26. The City submitted a revised *Engineering Report for the Vista Canyon Water Factory (Municipal Wastewater Treatment Facility)*, dated November 27, 2019, on its proposed production, distribution, and use of recycled water for irrigation as required by section 60323 of Title 22, CCR. On February 19, 2020, the revised Title 22 Engineering Report was conditionally approved by the DDW. This Order incorporates conditions and requirements consistent with DDW's recommendations in Attachment A. DDW's approval and acceptance conditions of the Title 22 Engineering Report are an enforceable part of this Order.
27. **State Water Board Resolution No. 77-1** – The State Water Board adopted Resolution No. 77-1, Policy with Respect to Water Recycling in California, which includes principles that encourage and recommend funding for water recycling and its use in water-short areas of the State.
28. **Recycled Water Policy** – State Water Board Resolution No. 2009-0011, *Adoption of a Policy for Water Quality Control for Recycled Water* (Recycled Water Policy) as amended in 2013 and 2018, is intended to support the State Water Board's Strategic Plan to promote sustainable local water supplies. Increasing the acceptance and promoting the use of recycled water is a means towards achieving sustainable local water supplies and can result in reduction in greenhouse gases, a significant driver of climate change. The Recycled Water Policy is also intended to encourage beneficial use of, rather than solely disposal of, recycled water generated from municipal wastewater sources in a manner that fully implements state and federal water quality laws.

The most recent amendment to the Recycled Water Policy was adopted on December 11, 2018 and became effective on April 8, 2019.<sup>1</sup> The amended Recycled Water Policy requires wastewater and recycled water dischargers to annually report monthly volumes of influent, wastewater produced, and effluent, including treatment level and discharge type. As applicable, dischargers are additionally required to annually report recycled water use by volume and category of reuse. This Order incorporates the reporting requirements of the Recycled Water Policy.

The Recycled Water Policy requires the development of salt and nutrient management plans (SNMPs) in basins where salts and/or nutrients are a threat to water quality and therefore planning is needed to achieve water quality objectives in the long term. SNMPs are required to include a monitoring program, source identification, assimilative capacity estimation, implementation measures to manage

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<sup>1</sup> Recycled Water Policy Amendment  
[https://www.waterboards.ca.gov/board\\_decisions/adopted\\_orders/resolutions/2018/121118\\_7\\_final\\_amendment\\_oal.pdf](https://www.waterboards.ca.gov/board_decisions/adopted_orders/resolutions/2018/121118_7_final_amendment_oal.pdf)

or reduce salt and nutrient loading in the basin on a sustainable basis, and an antidegradation analysis.

In accordance with the Recycled Water Policy, the Upper Santa Clara River Integrated Regional Water Management Group, which is comprised of the Castaic Lake Water Agency (CLWA) (currently SCVWA), the City, Santa Clarita Water Division (currently SCVWA-Santa Clarita Division), Los Angeles County Flood Control District, Newhall County Water District (currently SCVWA-Newhall Division), San Gabriel & Lower Los Angeles Rivers and Mountains Conservancy, SCVSD, and Valencia Water Company (currently SCVWA-Valencia Division) entered into a Memorandum of Understanding to prepare the *Salt and Nutrient Management Plan Santa Clara River Valley East Subbasin (SNMP East Subbasin)*. This group of agencies, collectively known as the Salt and Nutrient Task Force and facilitated by the CLWA, directed the preparation of the *SNMP East Subbasin* and the accompanying substitute environmental document, which was prepared using guidance set forth by the Regional Water Board. The Regional Water Board amended the Basin Plan to incorporate management measures from the *SNMP East Subbasin* under the title, “a Program of Implementation Consisting of Groundwater Quality Management Measures for Salt and Nutrients in the Upper Santa Clara River Basin,” Resolution No. R16-008, on December 8, 2016. The Office of Administrative Law approved the Basin Plan amendment on June 19, 2018.

29. **Antidegradation Policy, State Water Board Resolution No. 68-16** – State Water Board Resolution 68-16 (“*Statement of Policy with Respect to Maintaining High Quality of Waters in California*”) requires that, whenever the existing quality of water is better than the quality established in policies as of the date on which such policies become effective, such existing high quality must be maintained. Resolution 68-16 only allows change in the existing high quality if it has been demonstrated to the Water Board that the change is consistent with maximum benefit to the people of the State, will not unreasonably affect present and anticipated beneficial uses of such water, and will not result in water quality less than that prescribed in the policies. Resolution 68-16 further requires that discharges meet WDRs which will result in the best practicable treatment or control of the discharge necessary to assure that (a) pollution or nuisance will not occur and (b) the highest water quality consistent with the maximum benefit to the people of the State will be maintained.

## **ANTIDEGRADATION ANALYSIS**

30. The Recycled Water Policy provides that, for non-potable recycled water projects within a groundwater basin with a basin plan amendment based on an accepted SNMP, compliance with the Antidegradation Policy may be based in part on the technical findings of the SNMP or basin plan amendment, as applicable. Therefore, the antidegradation analysis for these WDRs/WRRs is based on the technical

findings of the *SNMP East Subbasin* and additional site-specific modeling for the Water Factory project.

The purpose of the *SNMP East Subbasin* is to document the current ambient water quality conditions in the East Subbasin and to ensure that all water management practices, including the use of recycled water, ensure protection of beneficial uses and allow for the sustainable use of groundwater resources, consistent with water quality objectives for the East Subbasin. The *SNMP East Subbasin* uses a numerical model to predict the impact of the projected discharge from the Water Factory, along with other projects in the East Subbasin, on salt and nutrient loading in the East Subbasin. The SNMP used the 50<sup>th</sup> percentile of ambient water quality data from potable wells between 2001 and 2011 to determine the assimilative capacity of the East Subbasin. Assimilative capacity is the difference between existing water quality and water quality objectives. The *SNMP East Subbasin* identifies six management zones. Management Zone 1 was separated into Zones 1a and 1b, and the Water Factory is located in Management Zone 1a. The *SNMP East Subbasin* identifies the Water Factory as an individual project. In accordance with the Recycled Water Policy and the Regional Water Board assistance document for the preparation of SNMPs, in order to demonstrate compliance with the Antidegradation Policy, an individual project that utilizes less than 10 percent of the available assimilative capacity (or multiple projects utilizing less than 20 percent of the available capacity in a basin/sub-basin) in a basin/sub-basin need only conduct an antidegradation analysis verifying the use of the assimilative capacity.

In 2016, the Discharger conducted a groundwater impact analysis using a model consistent with the SNMP to evaluate the potential localized impact to groundwater from the Water Factory discharge in Management Zone 1a underlying the recycled water application area. The groundwater impact analysis was reviewed and approved by the Regional Water Board and reflected in the previous Order No. R4-2016-0220.

In 2019, the Discharger conducted another groundwater impact analysis based on more recent ambient groundwater data and revised projected effluent quality. The additional groundwater data included concentrations of TDS, sulfate, chloride, boron and nitrate (as nitrogen) collected by the SCVWA from 15 active water supply wells (Pinetree 1, Pinetree 3, Pinetree 4, Pinetree 5, Honby, Lost Canyon 2, Lost Canyon 2A, Mitchell 5A, Mitchell 5B, North Oaks Central, North Oaks East, North Oaks West, Sand Canyon, Santa Clara, and Sierra) in Zone 1a. The City analyzed all available groundwater data collected from the 15 groundwater wells between 2001 and 2019 to determine the site-specific background groundwater quality. Consistent with the methodology used in the SNMP, the 50<sup>th</sup> percentile of the groundwater quality data was used to represent the ambient groundwater quality of Zone 1a (Table 5).

**Table 5. 50<sup>th</sup> Percentile Ambient Groundwater Quality in the Vista Canyon Project Area (Management Zone 1a)**

| Condition | TDS<br>(mg/L) | Chloride<br>(mg/L) | Sulfate<br>(mg/L) | Boron<br>(mg/L) | Nitrate as nitrogen<br>(mg/L) |
|-----------|---------------|--------------------|-------------------|-----------------|-------------------------------|
| 2001-2019 | 747.5         | 101                | 121               | 0.92            | 3.61                          |

Based on this data analysis, the ambient groundwater quality in the Water Factory area (Zone 1a) is better than the Basin Plan water quality objectives (Table 3). Therefore, the receiving water limits for groundwater in this Order are based on the ambient groundwater quality and, consistent with the Antidegradation Policy and Recycled Water Policy, the addition of 10 percent of the difference between the Basin Plan groundwater quality objectives and the 50<sup>th</sup> percentile of the ambient groundwater quality.

The antidegradation analysis must also show that the Water Factory effluent quality will not cause an exceedance of the receiving water limitations set to comply with the Antidegradation Policy and Recycled Water Policy.

**A. Water Factory Influent Quality**

- a. Because the Water Factory is a new facility, its effluent quality must be estimated based on the influent quality and the effects of the facility's treatment process on the influent quality. The Water Factory is a tertiary treatment facility and its treatment process removes many constituents from the influent. However, the Water Factory is not designed to treat salts, such as TDS, chloride, sulfate, and boron, in the influent. The influent quality is determined based on the potable water supply that will be delivered to the Vista Canyon Project and the addition of salts by users of the water before it is conveyed to the Water Factory. The Santa Clarita Division, one of the divisions under the SCVWA (formerly Santa Clarita Water Division), is the primary water agency that supplies potable water to the Vista Canyon Project. The SCVWA (Newhall Division, Santa Clarita Division, and Valencia Division) and Los Angeles Waterworks District No. 36 are water suppliers delivering potable water to communities in the Santa Clarita Valley.
- b. On March 14, 2019, the Water Factory submitted a technical memorandum that reviewed the monthly potable water data collected between January 2017 and December 2018 by Santa Clarita Water Division (SCWD) from a potable water turnout. The memorandum was updated on October 5, 2019 to include more water supply data through August 2019, which would closely represent the projected influent concentrations of TDS, sulfate, chloride, and boron (collectively salts). The monthly potable water data collected by SCWD indicated that salt

concentrations in potable water vary seasonally and are typically elevated in summer months. Table 6 (Column No. 2) shows the 95<sup>th</sup> percentile concentration of the monthly potable water supply data from January 2017 to August 2019, except for sulfate. Since sulfate data are not available, a sulfate concentration of 150 mg/L is assumed based on groundwater quality objectives in the Basin Plan.

- c. The estimated influent quality of the Water Factory (Column 4 of Table 6) results from the summation of Column 2 (95<sup>th</sup> percentile of sampled potable water quality) and Column 3 in Table 6. Column 3 in Table 6 is the addition of pollutant concentrations as the result of contributions from household and commercial use, adopted from the *Engineering Report for the Vista Canyon Water Factory (Municipal Wastewater Treatment Facility)*, dated November 27, 2019.

**Table 6. Projection of Water Factory Influent Water Quality**

| 1           | 2                                                   | 3                                | 4                                       |
|-------------|-----------------------------------------------------|----------------------------------|-----------------------------------------|
| Constituent | 95 <sup>th</sup> Percentile of Potable Water (mg/L) | Addition to Potable Water (mg/L) | Projected Water Factory Influent (mg/L) |
| TDS         | 568                                                 | 225                              | 793                                     |
| Chloride    | 96                                                  | 31                               | 127                                     |
| Sulfate     | 150                                                 | 20                               | 170                                     |
| Boron       | 0.90                                                | 0.15                             | 1.05                                    |
| Nitrate-N   | 2.87                                                | Not available                    | Varied                                  |

- d. The City Plumbing Code adopted on November 26, 2013 and the SCVSD Ordinance adopted on June 11, 2008 prohibit water softener installation within the Vista Canyon Project. The purpose of the Code and the Ordinance is to limit the discharge of TDS, including chloride, to the Water Factory.

#### B. Water Factory Effluent Quality

The Water Factory is a tertiary treatment facility and its treatment process removes many constituents from the influent. For example, the projected Ammonia-N + Nitrate-N + Nitrite-N concentration in the effluent is based on expected performance of the full nitrification and denitrification (NDN) process at the Water Factory. However, as noted above, the Water Factory is not designed to treat salts such as TDS, chloride, sulfate, and boron in the influent. Therefore, the salt concentrations in the effluent will be the same as those in



the influent presented in Table 6. Based on the tertiary treatment process, recent projections of influent water quality, and assuming proper operation and maintenance, the Water Factory will achieve the effluent concentrations detailed in Table 7 below:

**Table 7. Projection of Water Factory Effluent Water Quality**

| Constituents                      | Units | Concentrations |
|-----------------------------------|-------|----------------|
| TDS                               | mg/L  | 793            |
| Chloride                          | mg/L  | 127            |
| Sulfate                           | mg/L  | 170            |
| Boron                             | mg/L  | 1.05           |
| Ammonia-N + Nitrate-N + Nitrite-N | mg/L  | 6.0            |

Table 7 notes: Nitrate as nitrogen in effluent is expected to be less than 6.0 mg/L with the full NDN process at the Water Factory.

The projected effluent concentrations of nitrate, TDS, sulfate, chloride, and boron shown in Table 7 are greater than the ambient groundwater quality based on the updated water supply data (January 2017 to August 2019) and groundwater quality data (February 2001 to May 2019). However, the updated 2019 groundwater impact analysis demonstrates, as did the 2016 analysis, that the Water Factory can discharge at the projected effluent quality and still attain the receiving water limitations, as discussed below.

Table 8 below provides the results of the groundwater impact analysis, presenting the Basin Plan groundwater quality objectives, the 50<sup>th</sup> percentile ambient groundwater quality, and the projected groundwater quality resulting from varying years of Water Factory discharge. The impact analysis used the 95<sup>th</sup> percentile of the concentrations of TDS, chloride sulfate, boron and nitrate as a constant discharge, and evaluated the corresponding groundwater quality after 1, 10, 20, 30 and 50 years. The model results indicate that the use of recycled water generated from the Water Factory will not cause exceedances of the receiving water limitations based on the Antidegradation Policy for TDS and chloride over a 50-year period of discharge. However, the assimilative capacity for sulfate and boron allocated to the Water Factory by the *SNMP East Subbasin*, which is equivalent to 10 percent of the difference between the Basin Plan groundwater quality objectives and the 50<sup>th</sup> percentile ambient groundwater quality may be depleted after approximately 20 and 50 years of discharge, respectively, but would still be below the groundwater quality objectives.

**Table 8. Antidegradation Analyses: Comparison of Projected Groundwater Quality, Ambient Groundwater Quality, and Basin Plan Groundwater Quality Objectives**

|                                                                             |                 | <b>TDS<br/>(mg/L)</b> | <b>Chloride<br/>(mg/L)</b> | <b>Sulfate<br/>(mg/L)</b> | <b>Boron<br/>(mg/L)</b> | <b>Nitrate as<br/>nitrogen<br/>(mg/L)</b> |
|-----------------------------------------------------------------------------|-----------------|-----------------------|----------------------------|---------------------------|-------------------------|-------------------------------------------|
| <b>Basin Plan Groundwater<br/>Quality Objective</b>                         |                 | 800                   | 150                        | 150                       | 1.0                     | 10                                        |
| <b>50<sup>th</sup> Percentile Ambient<br/>Groundwater Quality</b>           |                 | 747.5                 | 101                        | 121                       | 0.92                    | 3.61                                      |
| <b>Receiving Water<br/>Limitations based on<br/>Anti-Degradation Policy</b> |                 | 752.8                 | 105.9                      | 123.9                     | 0.93                    | 4.25                                      |
| <b>Projected<br/>Groundwater<br/>Quality</b>                                | <b>1 Year</b>   | 747.6                 | 101.1                      | 121.1                     | 0.92                    | 3.62                                      |
|                                                                             | <b>10 Years</b> | 748.8                 | 101.7                      | 122.4                     | 0.92                    | 3.68                                      |
|                                                                             | <b>20 Years</b> | 750.1                 | 102.5                      | 123.8                     | 0.93                    | 3.75                                      |
|                                                                             | <b>30 Years</b> | 751.3                 | 103.2                      | 125.1                     | 0.93                    | 3.81                                      |
|                                                                             | <b>50 Years</b> | 753.7                 | 104.5                      | 127.6                     | 0.94                    | 3.94                                      |

Table 8 notes: The 50<sup>th</sup> percentile concentrations for ambient groundwater quality are based on data collected from 15 wells between 2001 and 2019, as shown in Table 5.

31. In summary, this Order establishes effluent and receiving water limitations that will comply with the Antidegradation Policy and Recycled Water Policy. The project is allowed to utilize 10% of the available assimilative capacity of the East Subbasin. The impact caused by sulfate and boron is localized to Management Zone 1a; the assimilative capacity of the East Subbasin as a whole will not be depleted. The application of the recycled water for irrigation purposes, allowed by this Order, will not unreasonably affect present and anticipated beneficial uses of the affected water bodies. Similarly, no pollution or nuisance is expected to occur and the highest water quality consistent with the maximum benefit to the people of the state will be maintained.

This Order will prevent unreasonable threats to present and anticipated beneficial uses and will not result in receiving groundwater quality that exceeds water quality objectives set forth in the Basin Plan. Limitations for each waste constituent are based on the most stringent applicable water quality objective to protect the most sensitive beneficial use of the receiving waterbody; thus, all beneficial uses are protected. To ensure such protection, a vigorous monitoring plan is also adopted with this Order. A discussion of the basis for each of the effluent and receiving water limitations is provided in Findings 32 and 33. This Order contains requirements for

ensuring that Best Practicable Technology and Control (BPTC) and the highest water quality consistent with the maximum benefit to the people of the State will be achieved. The Water Factory uses UV disinfection, which reduces the amount of chlorine, and therefore salinity in the effluent, to achieve the required disinfection compared to chlorination alone. Accordingly, the discharge is consistent with the antidegradation provisions of Resolution 68-16. Based on the results of wastewater treatment and monitoring of effluent and groundwater quality, the Regional Water Board may reopen this Order to reconsider groundwater limitations and other requirements to comply with Resolution 68-16, if necessary.

## **RATIONALE FOR LIMITATIONS**

32. The numeric receiving water limitations for groundwater quality imposed by this Order are based on the following.
  - A. TDS, chloride, sulfate, boron, and nitrogen compounds (Ammonia-N + Nitrate-N + Nitrite-N) – Groundwater limitations are based on protecting background groundwater quality set at the values presented in Table 8, *supra*, and in consideration of the antidegradation policy, by adding 10 percent of the difference between the Basin Plan Groundwater Quality Objective and the 50<sup>th</sup> percentile of the ambient groundwater quality.
  - B. Nitrite as nitrogen, total coliform, fecal coliform, and enterococcus – Groundwater limitations are based on the water quality objectives for ground waters in the Basin.
33. The numeric effluent limitations imposed by this Order are based on the following:
  - A. Oil and grease – Effluent limitations are based on the narrative water quality objective for surface waters in the Basin Plan and on best professional judgment to interpret the narrative objective. The same numeric limitations are specified in other permits for tertiary-treated wastewater treatment plants adopted by this Regional Water Board.
  - B. Biochemical oxygen demand and total suspended solids – Effluent limitations are technology-based limits contained in similar orders for Publicly Owned Treatment Works (POTWs) indicative of treatment levels that are achievable by tertiary-treated wastewater treatment systems.
  - C. Methylene blue activated substances – Effluent limitations are based on the water quality objective in the Basin Plan consistent with Title 22 CCR secondary MCLs.
  - D. Nitrogen compounds – Effluent limitations are based on the projected Water Factory effluent water quality shown in Table 7, which will ensure attainment of

receiving water limitations based on the groundwater impact analysis as shown in Table 8.

- E. TDS and chloride – Effluent limitations are based on projected Water Factory effluent water quality presented in Tables 6 and 7, which will ensure attainment of receiving water limitations based on the groundwater impact analysis as shown in Table 8.
- F. Sulfate and boron – Effluent limitations are based on the Basin Plan groundwater quality objectives.
- G. The Regional Water Board has determined that it is not practicable to express effluent limitations for TDS, chloride, sulfate, and boron as daily maximum, average weekly, and average monthly effluent limitations because of seasonal variations in the influent source water quality. Further, groundwater quality objectives in the Basin Plan were established primarily to protect the underlying groundwater and impact on groundwater quality from the discharge of these pollutants to land is not immediate. Consequently, a 12-month averaging period is more appropriate. To encourage beneficial reuse of treated wastewater consistent with the State Water Board policy (Finding 27), Section II of this Order incorporates provisions that allow for effluent limitations based on a flow weighted 12-month rolling average period.

## **CEQA AND NOTIFICATION**

- 34. The City is the lead agency for purposes of the California Environmental Quality Act (CEQA) (Public Resources Code section 21000 et seq). The City released a Notice of Preparation (NOP) on October 1, 2009. The NOP provided notice to the public and public agencies that an Environmental Impact Report (EIR) would be prepared for the construction of the Vista Canyon Project. The Draft EIR was released for public comment on October 19, 2010, with notices published in the Signal Newspaper, notices mailed to interested parties, and notices mailed to the State Clearinghouse for circulation to responsible agencies (SCH No. 2007071039). The comment was due on December 3, 2010. Thirty-one written and oral comments were received, including a comment letter from the Regional Water Board dated December 2, 2010. The City conducted Planning Commission Meetings on October 19, 2010, November 2, 2010, and December 21, 2010 and a City Council Hearing on March 22, 2011 to accept verbal comments on the Draft EIR. On April 26, 2011, the City Council held a public hearing and certified the Final EIR.

The EIR had identified the potential impacts on wastewater disposal and water quality, resulting from the development of the Vista Canyon Project. To mitigate the impacts to groundwater and surface water quality caused by wastewater disposal, the Water Factory Project, pursuant to local, regional, state and federal design

standards, proposed to treat the domestic wastewater to the Title 22 Recycled Water standards at the Water Factory. The discharge is also required to obtain all necessary permits for the construction of the Vista Canyon Project and the Water Factory Project.

35. The Regional Water Board is a responsible agency for purposes of CEQA and has considered the EIR prepared by the City. The Regional Water Board has incorporated requirements into this Order to protect the quality of the waters of the state consistent with the applicable plans and policies that apply to the discharges regulated by this Order. This Order is consistent with the mitigation measures identified by the City in the EIR because it requires compliance with Title 22 standards. This Order includes a monitoring and reporting program to determine compliance with the terms of the Order, including the Title 22 standards, and to ensure protection of water quality. The Regional Water Board finds that all environmental effects have been identified for project activities that it is required to approve, and that the Project will not have significant adverse impacts on the environment provided that the mitigation presented in the final EIR and conditions of this Order are carried out. In adopting this Order, the Regional Water Board has eliminated or substantially lessened the less-than-significant effects on water quality, and therefore approves the project.
36. **Petition** – Any person aggrieved by this action of the Regional Water Board may petition the State Water Board to review the action in accordance with CWC section 13320 and CCR Title 23, sections 2050 and following. The State Water Board must receive the petition by 5:00 p.m., 30 days after the date of this Order, except that if the thirtieth day following the date of this Order falls on a Saturday, Sunday, or a state holiday, the petition must be received by the State Water Board by 5:00 pm on the next business day. Copies of the law and regulations applicable to filing petitions may be found on the Internet at [http://www.waterboards.ca.gov/public\\_notices/petitions/water\\_quality/](http://www.waterboards.ca.gov/public_notices/petitions/water_quality/) or will be provided upon request.
37. **Public Notice** – On February 14, 2020, the Regional Water Board notified the City and interested agencies and persons of its intent to issue WDRs/WRRs in Order No. R4-2020-0049 to revise and replace WDRs/WRRs in Order No. R4-2016-0220 with updated treatment processes, and effluent and groundwater limitations for the distribution and use of tertiary-treated and disinfected effluent as recycled water. The Regional Water Board has provided the City and interested agencies and persons with an opportunity to submit written comments.

The Regional Water Board, in a public meeting, heard and considered all comments pertaining to these WDRs/WRRs.

**IT IS HEREBY ORDERED** that the City shall comply with the following:

**I. INFLUENT LIMITS AND REQUIREMENTS**

Influent waste shall be limited to domestic wastewater from the Vista Canyon Project and a portion of existing domestic flows from the City and shall not exceed its design capacity of 392,000 GPD.

**II. TERTIARY-TREATED EFFLUENT/RECYCLED WATER LIMITATIONS AND REQUIREMENTS**

1. Tertiary-treated effluent/recycled water discharged from the Water Factory shall not exceed 371,000 GPD.
2. Tertiary-treated effluent/recycled water shall not contain constituents with concentrations exceeding limits listed in Tables 9 and 10 and Sections II.3 through II.5. Compliance with the effluent limitations will be determined at the location indicated as “Recycled Water Flow Meter & Sampling” in Figure 4.

**Table 9. Effluent/Recycled Water Limitations for Oil and Grease, TSS, BOD, MBAS, and Nitrogen Compounds**

| Constituents                                       | Units     | Monthly Average | Daily Maximum |
|----------------------------------------------------|-----------|-----------------|---------------|
| Oil and grease                                     | mg/L      | 10              | 15            |
| Total suspended solids                             | % removal | ≥ 85            |               |
|                                                    | mg/L      | 15              | 45            |
| Biochemical oxygen demand (BOD <sub>5@20°C</sub> ) | % removal | ≥ 85            |               |
|                                                    | mg/L      | 20              | 45            |
| Methylene blue activated substances (MBAS)         | mg/L      | 0.5             |               |
| Ammonia-N + Nitrate-N + Nitrite-N                  | mg/L      |                 | 6.0           |

**Table 10. Effluent/Recycled Water Limitations for TDS, Chloride, Sulfate, and Boron**

| Constituents | Units | Flow Weighted 12-month Rolling Average |
|--------------|-------|----------------------------------------|
| TDS          | mg/L  | 793                                    |
| Chloride     | mg/L  | 127                                    |
| Sulfate      | mg/L  | 150                                    |
| Boron        | mg/L  | 1.0                                    |

- A. The flow weighted 12-month rolling average concentration of TDS, Chloride, Sulfate, and Boron shall not exceed the effluent limitations specified in Table 10 as determined in section IX.
  - B. The Discharger shall monitor and report the quality of potable water supply and make all reasonable efforts to improve the quality of the potable water supply in the Discharger's service area and thereby the wastewater. Examples of such efforts include participation in regional salinity control and stormwater infiltration projects.
  - C. If the flow weighted 12-month rolling average effluent concentration for a constituent exceeds 95% of the effluent limitation concentration for 3 consecutive months, the Discharger shall, within 60 days, submit to the Regional Water Board a plan and time schedule for implementation of measures to ensure that the 12-month flow weighted rolling average effluent limitation is not exceeded. Such measures include discharging to Valencia WRP. The plan and schedule are to be implemented upon Regional Water Board approval.
3. The pH of tertiary-treated effluent/recycled water shall at all times be within the range of 6.5 to 8.5. Excursion of this range shall not be considered a violation provided the duration is not more than 10 minutes in a 24-hour period, and pH shall at all times be within 6 and 9.
4. The tertiary-treated effluent/recycled water shall be filtered and subsequently disinfected with UV that meets the following criteria:
- A. UV disinfection shall comply with the "*Ultraviolet Disinfection Guidelines for Drinking Water and Water Reuse*" published by National Water Research Institute in August 2012, which specifies for permeability of membrane filtration that:
    - a. The design UV dose shall deliver a minimum equivalent dose of 100 millijoules per square centimeter ( $\text{mJ}/\text{cm}^2$ ) at all times; and
    - b. The filtered effluent UV transmittance shall be 55%.
  - B. The City shall submit a performance testing protocol for the UV system prior to operation and submit results of the performance testing to the Executive Officer of the Regional Water Board and DDW prior to the initial discharge.
  - C. In accordance with Title 22 CCR section 60301.230, effluent shall be, at all times, adequately disinfected and oxidized. In the event that the effluent exceeds any of the following, based on daily grab samples, the

City shall suspend recycled water applications until such time that the cause of the failure has been identified and corrected. Any failure to meet the total coliform limits shall be reported to the DDW and the Regional Water Board in the next quarterly report.

- a. A 7-day median of 2.2 most probable number (MPN) per 100 milliliters for two consecutive days;
  - b. 23 MPN per 100 milliliters in more than one sample in any 30-day period; and
  - c. 240 MPN per 100 milliliters in any sample.
- D. In accordance with Title 22 CCR section 60301.320, filtered wastewater shall be an oxidized wastewater that has been coagulated and passed through a bed of filter media under the following conditions:
- a. At a rate that does not exceed 5 gallons per minute per square foot of surface area in mono, dual or mixed media gravity upflow or pressure filtration systems, or does not exceed 2 gallons per minute per square foot of surface area in a traveling bridge automatic backwash filter; and
  - b. The turbidity of the filtered wastewater does not exceed any of the following:
    - i. An average of 2 Nephelometric Turbidity Units (NTU) within a 24-hour period;
    - ii. 5 NTU more than 5 percent of the time within a 24-hour period; and
    - iii. 10 NTU at any time.
5. Maximum Contaminant Limits: The effluent shall not contain trace, toxic and other constituents in concentrations exceeding the applicable maximum contaminant levels (Attachment B) for drinking water established by the DDW in sections 64431 (Attachment B-1), 64442 (Attachment B-2), 64443 (Attachment B-3), 64444 (Attachment B-4), 64449 (Attachment B-5), and 64533 (Attachment B-6), Article 5, Chapter 15, Title 22 of the CCR, or subsequent revisions, or at levels that adversely affect the beneficial uses of receiving groundwater. Concentrations of contaminants in the effluent shall, at all times, not exceed the following MCLs. In case of a violation of any primary or secondary MCL, the City shall notify and submit a report according to Provision X.5. of this Order.



- A. Primary MCLs specified in Chapter 15, Domestic Water Quality and Monitoring, Title 22, CCR:
  - a. Inorganic chemicals in section 64431, Table 64431-A, except for nitrogen compounds, Attachment B-1 of this Order;
  - b. Radionuclides in section 64442, Table 64442, Attachment B-2 and section 64443, Table 64443, Attachment B-3 of this Order; and
  - c. Regulated organic chemicals in section 64444, Table 64444-A, Attachment B-4 of this Order.
- B. Secondary MCLs in Chapter 15, Domestic Water Quality and Monitoring, Title 22, CCR, Table 64449-A, Attachment B-5 of this Order.
- C. Primary MCLs for disinfection byproducts specified in Chapter 15.5, Article 2, section 64533, Table 64533-A, Attachment B-6 of this Order.

### III. RECEIVING WATER LIMITATIONS FOR GROUNDWATER QUALITY

- 1. The City is prohibited from altering the quality or elevation of the underlying groundwater.
- 2. Groundwater shall not contain constituents with concentrations exceeding limits specified in Attachments B-1 to B-6 and Table 11 as a result of this discharge. Compliance with these limits will be determined at groundwater wells MW-1, MW-2, MW-3, MW-4, MW-5, MW-6, Sand Canyon, Mitchell 5B, and Sierra. More information of these groundwater monitoring wells is available in Table 4, section V.3.A of the accompanying MRP No. CI-10041.

**Table 11. Receiving Water Limitations for Groundwater Quality**

| Constituents                      | Units     | Single Sample Maximum |
|-----------------------------------|-----------|-----------------------|
| TDS                               | mg/L      | 753                   |
| Chloride                          | mg/L      | 106                   |
| Sulfate                           | mg/L      | 124                   |
| Boron                             | mg/L      | 0.93                  |
| Ammonia-N + Nitrate-N + Nitrite-N | mg/L      | 4.2                   |
| Nitrite-N                         | mg/L      | 1.0                   |
| Total coliform                    | MPN/100mL | 1.1                   |
| Fecal coliform                    | MPN/100mL | 1.1                   |

| Constituents | Units     | Single Sample Maximum |
|--------------|-----------|-----------------------|
| Enterococcus | MPN/100mL | 1.1                   |

Table notes: The unit of MPN/100mL denotes most probable number per 100 milliliters.

3. The City shall demonstrate that the discharge and recycled water use from the Water Factory do not contribute to the degradation of groundwater quality by meeting all groundwater quality limitations specified in Table 11 and Attachments B-1 to B-6. In the event that the groundwater quality exceeds the limitations specified in Table 11 and Attachments B-1 to B-6, the Discharger shall be in violation of this Order unless it can demonstrate that the discharge or recycled water use does not contribute to the exceedance of the groundwater quality limitations.
4. In addition to being a violation of this Order, an exceedance of the TDS, chloride, sulfate, boron, or nitrogen limitations in Table 11 constitutes an exceedance of allowable project-specific assimilative capacity allocated to the Water Factory in the *SNMP East Subbasin*. Consistent with the SNMP, the City shall submit a plan with a proposal to implement facilities and programs designed to offset the amount of salt discharged in excess of the receiving water limitations into the Santa Clara-Mint Canyon Subbasin. The plan shall be submitted within 90 days of the exceedance.

#### **IV. SPECIFICATIONS FOR PRODUCTION, OPERATION, AND USE OF RECYCLED WATER AND ITS FACILITIES**

1. The City is responsible for ensuring that appropriate ordinances are established to regulate production, operation, and use of recycled water and its facilities within the Vista Canyon Project and areas that overlie Management Zone 1A of the Santa Clara-Mint Canyon Subbasin, consistent with this Order.
2. The City shall submit a revised Title 22 Engineering Report to DDW and the Regional Water Board for review and approval if additional recycled water use is proposed.
3. Recycled water shall not be used for direct human consumption or for the processing of food or drink intended for human consumption.
4. The City shall receive final acceptance from DDW that Title 22 Engineering Report requirements and conditions specified in the DDW's conditional approval letter dated February 19, 2020 prior to the initial discharge of recycled water. The City is required to comply with all conditions specified in the Title 22 Engineering Report and the DDW's acceptance letter for the recycled water applications.

5. The delivery of recycled water to new end-users shall be subject to DDW and Regional Water Board approval and/or its delegated local agency.
6. Any effluent not meeting conditions specified in this Order including DDW's conditional acceptance letter in Attachment A shall be diverted to either the Saugus or Valencia WRP.

## **V. USE AREA REQUIREMENTS**

"Use area" means an area with defined boundaries, which may contain one or more facilities where recycled water is used. The City shall be responsible for ensuring that all users of recycled water comply with the following:

1. No irrigation with disinfected tertiary-treated recycled water shall take place within 50 feet of any domestic water supply well.
2. There shall be no storage or impoundment of disinfected tertiary-treated water within 100 feet of any domestic water supply well.
3. Recycled water shall be applied at such a rate and volume as not to exceed vegetative demand and soil moisture conditions. Special precautions must be taken to prevent clogging of spray nozzles and over-watering, and minimize the production of runoff. Pipelines shall be maintained so as to prevent leakage.
4. All above ground irrigation appurtenances need to be marked appropriately.
5. Any incidental runoff from recycled water projects shall be handled as follows:
  - A. The discharge of recycled water to surface water is prohibited.
  - B. Discharges of recycled water to surface waters may only occur where regulated under a separate National Pollutant Discharge Elimination System (NPDES) permit issued by the Regional Water Board.

Incidental runoff is defined as unintended small amounts (volume) of runoff from recycled water use areas, such as unintended, minimal over-spray from sprinklers that escapes the recycled water use area. Irrigation system maintenance shall be consistent with the requirements found in the State Water Board's Recycled Water Policy.

6. Spray, mist, or runoff shall not enter dwellings, designated outdoor eating areas, or food handling facilities, and shall not contact any drinking water fountain.
7. Recycled water shall not be used for irrigation during periods of rainfall and/or runoff.

8. Recycled water shall be retained on the designated area and shall not be allowed to escape as surface flow.
9. All recycled water use areas that are accessible to the public shall be posted with signs that are visible to the public, in a size no less than 4 inches high by 8 inches wide, that include the following wording: "RECYCLED WATER – DO NOT DRINK" as shown in Figure 8. Each sign shall display an international symbol similar to that shown in Figure 8. An alternative signage and wording may be used upon approval by the Executive Officer of the Regional Water Board.
10. No physical connection shall be made or allowed to exist between any recycled water piping and any piping conveying potable water, except as allowed under section 7604 of Title 17, CCR.
11. The portions of the recycled water piping system that are in areas subject to access by the general public shall not include any hose bibs (a faucet or similar device to which a common garden hose can be readily attached). Only quick couplers that differ from those used on the potable water system shall be used on the portions of the recycled water piping system in areas subject to public access.
12. Recycled water use shall not result in earth movement in geologically unstable areas.
13. The City or its authorized agency shall develop the User Agreements and Ordinances with the potential non-potable users of recycled water. Copies of the User Agreements and Ordinances shall be provided to the Regional Water Board and the DDW.
14. Use site-supervisors shall be appointed for the recycled water use areas and their staff shall be trained on the hazards of working with recycled water and periodically retrained.
15. For each new recycled water use area, the City shall provide the Regional Water Board and DDW with a description of the use area including, but not limited to: a description of the recycled water use (e.g. landscape, specific food crop, cooling tower, etc.); method of use (e.g. spray, flood, or drip); the location of domestic water supply facilities adjacent to the use areas; site containment measures; the party responsible for the distribution and use of the recycled water at the site; identification of other governmental entities which may have regulatory jurisdiction over the reuse site(s) such as State Food and Drug, State Licensing and Certification, County Health Department, etc. These Agencies

shall also be provided with a copy of the approved Title 22 Engineering Report for review and comment.

## **VI. REQUIREMENTS FOR DUAL-PLUMBED SYSTEMS**

1. “Dual plumbed” means a system that utilizes separated piping systems for recycled water and potable water within a facility and where the recycled water is used for either of the following purposes:
  - A. To serve plumbing outlets (excluding fire suppression systems) within a building; or
  - B. Outdoor landscape irrigation at individual residences.
2. The public water supply shall not be used as a backup or supplemental source of water for a dual-plumbed recycled water system unless the connection between the two (2) systems is protected by an air gap separation which complies with the requirements of section 7602 (a) and 7603 (a) of Title 17, CCR, and that such connection has been approved by the DDW and/or its delegated local agency.
3. The City or its authorized agency shall not deliver recycled water to a facility using a dual-plumbed system unless the report required pursuant to section 13522.5 of the CWC, which meets the requirements set forth in sections VI.4. and/or VI.5. of this Order, has been submitted and approved by DDW or its delegated local agency and the Regional Water Board. The Regional Water Board shall be furnished with a copy of the DDW approval within 30 days following the approval.
4. Prior to the initial operation of the dual-plumbed recycled water system and annually thereafter, the dual-plumbed system within each facility and use site shall be inspected by the City or its authorized agency for possible cross connections with the potable water system. The recycled water system shall also be tested for possible cross connections at least once every four (4) years. The inspections and the shutdown testing shall be performed by a cross connection control specialist certified by the California-Nevada section of the American Water Works Association or an organization with equivalent certification requirements. A written report documenting the result of the inspection and shutdown testing for the prior year shall be submitted to the DDW and the Regional Water Board within 30 days following completion of the inspection or shutdown testing. The procedures used to conduct the shutdown testing must be described.

5. The City shall notify DDW of any incidence of backflow from the dual-plumbed recycled water system into the potable water system within 24 hours of discovery of the incident.
6. Any backflow prevention device installed to protect the public water system serving the dual-plumbed recycled water system shall be inspected and maintained in accordance with section 7605 of Title 17, CCR.
7. The City shall obtain final approval from DDW prior to the initial use of recycled water.

## **VII. GENERAL REQUIREMENTS**

1. Dischargers shall operate and maintain facilities, treatment operations, associated collection systems and outfalls in ways to preclude adverse impacts to surface water or groundwater from impacts predicted to occur due to climate change.
2. The Discharger shall evaluate the need to revise its Climate Change Effects Vulnerability Assessment and Management Plan (Climate Change Plan) every 3 years or more frequently when conditions change that may impact water quality. The Discharger shall submit a report of its evaluation or a revised Climate Change Plan based on the evaluation. Submittal of the Climate Change Plan is required pursuant to CWC section 13267. As required by this provision, a regional water board may require a person to submit technical or monitoring program reports which the regional water board requires. The Climate Change Plan is needed in order to assess and manage climate change related-effects associated with Discharger operations that may affect water quality.

The Climate Change Plan shall include an assessment of short and long term vulnerabilities of the facility(ies) and operations as well as plans to vulnerabilities of collection systems, facilities, treatment systems, and outfalls for predicted impacts in order to ensure that facility operations are not disrupted, compliance with permit conditions is achieved, and receiving waters are not adversely impacted by discharges. Control measures shall include, but are not limited to, emergency procedures, contingency plans, alarm/notification systems, training, backup power and equipment, and the need for planned mitigations to ameliorate climate-induced impacts including, but not limited to, changing influent and receiving water quality and conditions, as well as the impact of rising sea level (where applicable) storm surges and back-to-back severe storms that are expected to become more frequent.

3. The recycling facility and areas where any potential pollutants are stored shall be adequately protected from inundation and damage by storm flows and run-off.
4. Adequate freeboard and/or protection shall be maintained in the recycled water storage tanks and process tanks to ensure that direct rainfall will not cause overtopping.
5. The wastewater treatment and use of recycled water shall not result in nuisance conditions caused by breeding of mosquitoes, gnats, midges, or other pests.
6. Odors of sewage origin shall not be perceivable any time outside the boundary of the treatment facility.
7. The City shall, at all times, properly operate and maintain all treatment facilities and control systems (and related appurtenances), which are installed or used by the City to achieve compliance with the conditions of this Order. Proper operation and maintenance include: effective performance, adequate funding, adequate operator staffing and training, and adequate laboratory and process controls (including appropriate quality assurance procedures).
8. Any wastes that do not meet the foregoing requirements shall be held in impervious containers and discharged at a legal point of disposal.
9. A copy of these requirements shall be maintained at the water recycling facility so as to be available at all times to operating personnel.
10. The distribution and irrigation systems shall be maintained and periodically inspected by the City or its authorized agency for proper maintenance and operation.
11. Pursuant to section 3860 of Title 23 CCR, the Water Factory Project shall meet the standard conditions specified in Conditions of Certification File No. 12-034.

## **VIII. PROHIBITIONS**

1. Recycled water shall not be used for direct human consumption or for the processing of food or drink intended for human consumption.
2. Wastes discharged and recycled water applications shall not contain tastes, odors, color, foaming, any materials, or other objectionable characteristics in concentrations that would:
  - A. Affect human, animal, and plant life;

- B. Cause nuisance or adversely affect the beneficial uses and quality of the receiving groundwater; or
  - C. Impact surface water that may be in hydraulic connection with groundwater.
- 3. Discharge of waste classified as 'hazardous', as defined in section 2521(a) of Title 23, CCR, section 2510 et seq., is prohibited. Discharge of waste classified as 'designated,' as defined in CWC section 13173, in a manner that causes violation of receiving water limitations, is prohibited.
- 4. The recycled water storage basin and storage tank shall not contain floating materials, including solids, foams or scum in concentrations that cause nuisance, adversely affect beneficial uses, or serve as a substrate for undesirable bacterial or algae growth or insect vectors.
- 5. There shall be no onsite disposal of sludge. Sludge-drying activities are allowed, but only as an intermediate treatment prior to offsite disposal. Any offsite disposal of wastewater or sludge shall be made only to a legal point of disposal. For purposes of this Order, a legal disposal site is one for which requirements have been established by a California Regional Water Board or comparable regulatory entity, and which is in full compliance therewith. Any wastewater or sludge handling shall be in such a manner as to prevent its reaching surface waters or watercourses.
- 6. Odors originating at this Water Factory shall not be perceivable beyond the limits of the property owned by the City.
- 7. No new connections of using recycled water may be made without notification to the Regional Water Board and DDW.
- 8. The discharge of waste shall not create a condition of pollution, contamination, or nuisance.
- 9. Bypass, discharge or overflow of untreated wastes, except as allowed by section VIII.10. of this Order, is prohibited.
- 10. Bypass (the intentional diversion of waste stream from any portion of a treatment facility) is prohibited. The Regional Water Board may take enforcement action against the City for bypass unless:
  - A. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage. (Severe property damage means substantial physical damage to property, damage to the treatment facilities that cause them to become inoperable, or substantial and permanent loss in the absence of a



bypass. Severe property damage does not mean economic loss caused by delays in production.)

- B. There were no feasible alternatives to bypass, such as the use of auxiliary treatment facilities, retention of untreated waste, or maintenance during normal periods of equipment down time. This condition is not satisfied if adequate back-up equipment shall have been installed in the exercise of reasonable engineering judgment to prevent a bypass that could occur during normal periods of equipment downtime or preventive maintenance.
  - C. The City must submit written notice at least 24 hours in advance of the need for a bypass to the Regional Water Board Executive Officer.
- 11. Any discharge of wastewater from the treatment system (including the wastewater collection system) at any point other than specifically described in this Order and except as provided for in section VIII.10 of this Order, is prohibited and constitutes a violation of this Order.
  - 12. Any discharge of effluent/recycled water at any point(s) other than designated recycled water use areas is prohibited and constitutes a violation of this Order.
  - 13. The discharge of effluent, including runoff, spray or droplets from the irrigation system, shall not occur outside the boundaries of the land application area.
  - 14. The discharge of waste to surface waters of the state or of the United States is prohibited.

## **IX. COMPLIANCE DETERMINATION**

- 1. Compliance with the effluent limitations contained in section II of this Order will be determined as specified below:
  - A. General

Compliance determinations shall be based on available analyses for the time interval associated with the effluent limitation. Where only one sample analysis is available in a specified time interval (e.g., monthly or weekly average), that sample shall serve to characterize the discharge for the entire interval. If quarterly sample results show noncompliance with the average monthly limit and that sample result is used or compliance determinations for each month of the quarter, then three separate violations of the average monthly limit shall be deemed to have occurred.
  - B. Average Monthly Effluent Limitation (AMEL)

If the average of daily discharges over a calendar month exceeds the AMEL for a given parameter in Table 9, this will represent a single violation, though the Discharger will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger will be considered out of compliance for that calendar month. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which no sample (daily discharge) is taken, no effluent compliance determination can be made for that calendar month.

C. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge exceeds the MDEL for a given parameter in Table 9, the Discharger will be considered out of compliance for that parameter for that 1 day only within the reporting period. For any 1 day during which no sample is taken, no effluent compliance determination can be made for that day.

D. Flow Weighted 12-Month Rolling Average Effluent Limitation (12-MRAEL)

Compliance with the flow weighted 12-month rolling average limitations under Section II.2 Table 10 shall be determined by the arithmetic mean of the last twelve months, calculated as the sum of monthly discharges (monthly volume multiplied by average monthly concentration) during the last twelve months divided by the sum of the monthly volumetric discharges during the twelve-month time period. Months with no discharge shall not be included in the 12-month running average calculations.

2. Compliance with the receiving water limitations contained in section III of this Order will be determined as specified below:

A. Instantaneous Maximum Receiving Water Limitation

If the analytical result of a single grab sample is higher than the instantaneous maximum receiving water limitation for a parameter in Table 11, the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum receiving water limitation would result in two instances of non-compliance with the instantaneous maximum receiving water limitation).

## **X. PROVISIONS**

1. The City shall submit plans for any change of the recycled water project to and obtain approval from DDW and the Regional Water Board. The American Water Works Association Guidelines for the Distribution of Non-Potable Water shall be followed, including installation of purple pipe, adequate signs, etc. As-built drawings shall show the final locations of the potable water, sewer, and recycled water pipelines; and indicate adequate separation between the recycled water and potable domestic water lines, which shall also be marked clearly or labeled using separate colors for identification. In addition, a copy of each application to DDW for a recycled water project shall be delivered to the Regional Water Board for inclusion in the administrative file.
2. If the recycled water system lateral pipelines are located on an easement contiguous to a homeowners private property and where there is a reasonable probability that an illegal or accidental connection to the recycled water line could be made, the City or its authorized agency shall provide a buffer zone or other necessary measures between the recycled water lines and the easement to prevent any illegal or accidental connection to the recycled water lines. If the City or its authorized agency does not feel it can maintain adequate control of the recycled water system pipelines, the pipelines will need to be relocated or a physical barrier needs to be installed to prevent this type of potential problem. The homeowners need to be educated on the use of recycled water in the area. The City or its authorized agency should specify a plan to interface with the homeowners as a part of the Rules of Service Agreement in an adjacent property awareness program.
3. The City or its authorized agency shall inspect the recycled water use areas on a periodic basis. A report of findings of the inspection shall be submitted to the City that will incorporate it with the quarterly report, specified in the MRP, to the DDW and the Regional Water Board.
4. The City shall file with the Regional Water Board, under penalty of perjury, annual and quarterly reports on self-monitoring work performed according to the detailed specifications contained in MRP attached hereto and incorporated herein by reference, as directed by the Executive Officer. The results of any monitoring done more frequently than required at the location and/or times specified in the MRP shall be reported to the Regional Water Board. The Discharger shall comply with all of the provisions and requirements of the MRP.
5. The City shall notify DDW and this Regional Water Board by telephone or electronic means within 24 hours of knowledge of any violations of recycled water use conditions, any adverse conditions as a result of the use of recycled water and any discharge exceeding the effluent limits prescribed in this Order

- from the Water Factory or/and the recycled water storage basin; written confirmation shall follow within 5 working days from date of notification, unless otherwise specified in this Order. The report shall include, but not limited to, the following information, as appropriate:
- A. Nature and extent of the violation;
  - B. Date and time: when the violation started, when compliance was achieved; and, when discharge was suspended and restored, as applicable;
  - C. Duration of violation;
  - D. Cause(s) of violation;
  - E. Corrective and/or remedial actions taken and/or will be taken with time schedule for implementation to prevent future violations; and
  - F. Impact of the violation.
6. The direct use of disinfected recycled water for irrigation and unpaved roadway dust control could affect the public health, safety, or welfare; requirements for such uses are therefore necessary in accordance with section 13523 of the CWC.
7. This Order does not exempt the City and its authorized agencies from compliance with any other laws, regulations, or ordinances which may be applicable; they do not legalize the recycling and use facilities; and they leave unaffected any further constraint on the use of recycled water at certain site(s) that may be contained in other statutes or required by other agencies.
8. This Order does not alleviate the responsibility of the City and its authorized agencies to obtain other necessary local, state, and federal permits to construct facilities necessary for compliance with this Order; nor does this Order prevent imposition of additional standards, requirements, or conditions by any other regulatory agency. Expansion of the recycled water distribution facility shall be contingent upon issuance of all necessary requirements and permits, including a conditional use permit.
9. After notice and opportunity for a hearing, this Order may be modified, revoked and reissued, or terminated for cause, that include, but is not limited to: failure to comply with any condition in this Order, endangerment of human health or environment resulting from the permitted activities in this Order, obtaining this Order by misrepresentation or failure to disclose all relevant facts, and

acquisition of new information which could have justified the application of different conditions if known at the time of Order adoption.

The filing of a request by the City for modification, revocation and reissuance, or termination of the Order; or a notification of planned changes or anticipated noncompliance does not stay any condition of this Order.

10. The City shall furnish, within a reasonable time, any information that the Regional Water Board may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order. The City shall also furnish the Regional Water Board, upon request, with copies of records required to be kept under this Order for at least three years.
11. In an enforcement action, it shall not be a defense for the City that it would have been necessary to halt or to reduce the permitted activity in order to maintain compliance with this Order. Upon reduction, loss, or failure of the treatment facility, the City shall, to the extent necessary to maintain compliance with this Order, control production or all discharges, or both, until the facility is restored or an alternative method of treatment is provided. This provision applies, for example, when the primary source of power of the treatment facility fails, is reduced, or is lost.
12. This Order includes "Standard Provisions Applicable to Waste Discharge Requirements" (Attachment C – Standard Provisions). In the event of conflict between provisions stated herein and the Standard Provisions, the provisions stated herein prevail.
13. This Order includes the WDRs/WRRs and the attached MRP (No. CI-10041). If there is any conflict among provisions stated in the MRP and these WDRs/WRRs, those provisions stated herein before prevail.
14. After a year of applying infiltrating recycled water into the use areas overlying the Management Zone 1A of the Santa Clara-Mint Canyon Subbasin, the City shall update the operation, maintenance, and monitoring plan (OMM Plan) and submit it to the Regional Water Board for review and approval, if there is any change to the original OMM Plan. The Water Factory shall be operated in accordance with the approved plan.

The OMM Plan shall cover critical operational parameters to include routine testing procedures for optimization of the UV dose for disinfection and reduction of light-sensitive contaminants, and all treatment processes, maintenance and calibration schedules for all monitoring equipment, process alarm set points, and response procedures for all alarms in each treatment process of the Water Factory, including criteria for diverting recycled water if water quality

- requirements are not met, start-up, emergency response and contingency plans. During the first year of operation of the Water Factory, all treatment processes shall be optimized to reduce contaminant levels. The results of these initial optimization efforts shall be incorporated into the updated OMM Plan. The OMM Plan shall include staffing levels with applicable certification levels for the Water Factory operations personnel. Significant changes in the operation of any of the treatment processes shall be reported to the DDW and the Regional Water Board. Changes in the approved OMM Plan must be approved by the DDW and the Regional Water Board prior to instituting changes.
15. For any material change or proposed change in character, location or volume of recycled water, or its uses, the City shall submit at least 120 days prior to the proposed change an engineering report or addendum to the existing engineering report to the Regional Water Board and DDW [pursuant to CWC, sections 13260(c) and 13522 and CCR, Title 22, section 60320.080] for approval.
  16. The City shall provide an Annual Report described in the MRP to this Regional Water Board.
  17. In order to limit the presence of constituents of concern specified in section II in the effluent including regulated and unregulated contaminants identified in Attachments B-1 to B-6 and Attachments D to E of the accompanying MRP, the City shall, for the purposes of protecting public health, ensure that its equipment and facilities for treatment and disposal operate at levels of peak performance.
  18. Spill Clean-Up Contingency Plan (SCP) Requirements – Within six (6) months prior to discharge, the City is required to submit a SCP, which describes the activities and protocols to address clean-up of spills, overflows, and bypasses of untreated or partially treated wastewater from the City's collection system or treatment facilities. At a minimum, this SCP shall include sections on spill clean-up and containment measures, public notification, and monitoring. The City shall review and amend this SCP as appropriate after each spill from the Water Factory or in the service area of the Water Factory. The City shall include a discussion in the annual summary report of any modifications to the SCP and the application of the SCP to all spills during the year.
  19. Construction, Operation, and Maintenance Requirements
    - A. The Water Factory subject to this Order shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to CCR, Title 23, division 3, chapter 26 (sections 13625 – 13633).

- B. The City shall maintain in good working order a sufficient alternate power source for operating the wastewater treatment and disposal facilities. All equipment shall be located to minimize failure due to moisture, liquid spray, flooding, and other physical phenomena. The alternate power source shall be designed to permit inspection and maintenance and shall provide for periodic testing. If such alternate power source is not in existence, the City shall halt, reduce, or otherwise control all discharges upon the reduction, loss, or failure of the primary source of power.
- C. The City shall provide standby or emergency power facilities and/or storage capacity or other means so that in the event of plant upset or outage due to power failure or other cause, discharge of raw or inadequately treated sewage does not occur.

## 20. Collection System Requirements

The State Water Board adopted General WDRs for Sanitary Sewer Systems, (WQ Order No. 2006-0003) on May 2, 2006 and amended the WDRs by Order No. WQ 2008-0002-EXEC and Order No. WQ 2013-0058-EXEC, to provide a consistent, statewide regulatory approach to address sanitary sewer collection systems. The General WDRs require public agencies that own or operate sanitary sewer systems to develop and implement sewer system management plans and report all sanitary sewer overflows (SSOs) to the State Water Board's online SSO database. The City's collection system is part of the system that is subject to WQ Order No. 2006-0003, as amended. As such, the City must properly operate and maintain its collection system (40 CFR part 122.41(e)). The City must report any non-compliance (40 CFR part 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 CFR part 122.41(d)). The Discharger is currently enrolled under the SSO General WDRs.

## 21. Spill Reporting Requirements

- A. **Initial Notification** – Although State and Regional Water Board staff do not have duties as first responders, this requirement is an appropriate mechanism to ensure that the agencies that do have first responder duties are notified in a timely manner in order to protect public health and beneficial uses. For certain spills, overflows and bypasses, the City shall make notifications as required below:
  - a. In accordance with the requirements of Health and Safety Code section 5411.5, the City shall provide notification to the local health officer or the director of environmental health with jurisdiction over the affected water body of any unauthorized release of sewage or

other waste that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but no later than two (2) hours after becoming aware of the release.

- b. In accordance with the requirements of CWC section 13271, the City shall provide notification to the California Emergency Management Agency (Cal EMA) of the release of reportable quantities of hazardous substances or sewage that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but not later than two (2) hours after becoming aware of the release. CCR, Title 23, section 2250, established 1,000 gallons or more as a reportable quantity of sewage. The phone number for reporting these releases to the Cal EMA is (800) 852-7550.
- c. The City shall notify the Regional Water Board of any unauthorized release of sewage from the Water Factory that causes, or probably will cause, a discharge to a water of the state as soon as possible, but not later than two (2) hours after becoming aware of the release. This initial notification does not need to be made if the City has notified Cal EMA and the local health officer or the director of environmental health with jurisdiction over the affected waterbody. The phone number for reporting these releases of sewage to the Regional Water Board is (213) 576-6683. The phone numbers for after hours and weekend reporting of releases of sewage to the Regional Water Board are (213) 305-2284 and (213) 305-2253.

At a minimum, the following information shall be provided to the Regional Water Board:

- i. The location, date, and time of the release;
- ii. The water body that may be impacted by the discharge;
- iii. An estimate of the amount of sewage or other waste released and the amount that reached the receiving water at the time of notification;
- iv. If ongoing, the estimated flow rate of the release at the time of the notification;
- v. The name, organization, phone number and email address of the reporting representative; and
- vi. A certification that the State Office of Emergency Services and the local health officer or directors of environmental health with



jurisdiction over the possibly affected water bodies have been notified of the discharge.

- B. **Monitoring** – For spills, overflows and bypasses reported under section X.21.A, the City shall monitor as required below:

To define the geographical extent of spill's impact, the City shall obtain grab samples (if feasible, accessible, and safe) for all spills, overflows or bypasses of any volume that reach any waters of the State (including surface and ground waters). The City shall analyze the samples for total and fecal coliform, *Escherichia coli* (*[E. coli]*, if a fecal coliform test shows positive), *enterococcus*, and relevant pollutants of concern, upstream and downstream of the point of entry of the spill (if feasible, accessible and safe). This monitoring shall be done on a daily basis from time the spill is known until the results of two (2) consecutive sets of bacteriological monitoring indicate the return to the background level or the County Department of Public Health authorizes cessation of monitoring.

- C. **Reporting** – The initial notification required under section X.21.A shall be followed by:

- a. As soon as possible, but not later than twenty-four (24) hours after becoming aware of an unauthorized discharge of sewage or other waste from its wastewater treatment plant to a water of the state, the City shall submit a statement to Regional Water Board staff via email. If the discharge is 1,000 gallons or more, this statement shall certify that Cal EMA has been notified of the discharge in accordance with CWC section 13271. The statement shall also certify that the local health officer or director of environmental health with jurisdiction over the affected water bodies has been notified of the discharge in accordance with Health and Safety Code section 5411.5. The statement shall also include at a minimum the following information:
  - i. Agency, Order No., and MRP No.;
  - ii. The location, date, and time of the discharge;
  - iii. The water body that received the discharge;
  - iv. A description of the level of treatment of the sewage or other waste discharged;
  - v. An initial estimate of the amount of sewage or other waste released and the amount that reached the impacted water body;

- vi. The Cal EMA control number and the date and time that notification of the incident was provided to Cal EMA; and
    - vii. The name of the local health officer or director of environmental health representative notified (if contacted directly); the date and time of notification; and the method of notification (e.g., phone, fax, email).
  - b. A written preliminary report shall be submitted to the Regional Water Board within five (5) working days after disclosure of the incident via the State Water Board GeoTracker database under Global ID WDR100016910. The final written report shall be included in the next quarterly monitoring report submitted to the GeoTracker database above. The written report shall document the information required in paragraph section X.21.D below, monitoring results and any other information required in provisions of the Standard Provisions document including corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences.
  - c. The City shall include a certification in the annual summary report (due according to the schedule in the accompanying MRP) that states that the sewer system emergency equipment, including alarm systems, backup pumps, standby power generators, and other critical emergency pump station components were maintained and tested in accordance with the City's preventive maintenance plan. Any deviations from or modifications to the Plan shall be discussed.
- D. **Records** – The City shall prepare and maintain a record of all spills, overflows or bypasses of raw or partially treated sewage from its collection system or Water Factory. This record shall be made available to the Regional Water Board upon request and a spill summary shall be included in the annual report, as required in the MRP No. CI-10041. The record shall contain:
- a. The date and time of each spill, overflow, or bypass;
  - b. The location of each spill, overflow, or bypass;
  - c. The estimated volume of each spill, overflow, or bypass including gross volume, amount recovered and amount not recovered, monitoring results as required by section X.21.B;
  - d. The cause of each spill, overflow, or bypass;

- e. Whether each spill, overflow, or bypass entered a receiving water and, if so, the name of the water body and whether it entered via storm drains or other man-made conveyances;
  - f. Any corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences; and
  - g. The mandatory information included in Sanitary Sewer Overflows (SSO) online reporting for finalizing and certifying the SSO report for each spill, overflow, or bypass under the General WDRs.
- E. **Activities Coordination** – The Regional Water Board expects that the City will coordinate their compliance activities for consistency and efficiency with other entities that have responsibilities to implement: (i) this WDRs/WRRs permit, and (ii) the General WDRs.
- F. **Consistency with General WDRs** – The requirements contained in this Order in sections X.18 (SCP Requirements), X.19 (Construction, Operation, and Maintenance Requirements), and X.21 (Spill Reporting Requirements) are intended to be consistent with the requirements of the General WDRs. The Regional Water Board recognizes that there may be some overlap between the WDRs/WRRs permit provisions and General WDRs requirements. The requirements of the General WDRs are considered the minimum thresholds (see Finding 11 of WQ Order No. 2006-0003). To encourage efficiency, the Regional Water Board will accept the documentation prepared by the City under the General WDRs for compliance purposes, as satisfying the requirements in sections X.18, X.19, and X.21 provided the more stringent provisions enumerated in this Order, have also been addressed.
22. The City shall submit to the Regional Water Board an Operation, Maintenance, and Monitoring Plan (OMM Plan) for the entire Water Factory and disposal facilities prior to startup of the Water Factory. The OMM Plan shall address all conditions specified in the DDW's approval letter. The City shall maintain the OMM Plan in useable condition, and available for reference and use by all applicable personnel. The City shall regularly review, and revise or update as necessary, the OMM Plan in order for the document(s) to remain useful and relevant to current equipment and operation practices. Reviews shall be conducted annually, and revisions or updates shall be completed as necessary and submitted to the Regional Water Board on an annual basis.

## **XI. REOPENER**

This Order may be reopened any time at the Regional Water Board's discretion to include the most scientifically relevant limitations or other requirements and may specifically be reopened to make revisions based on monitoring or to be consistent with revisions to the *SNMP East Subbasin*.

## **XII. EFFECTIVE DATE OF THE ORDER**

This Order takes effect upon its adoption.

I, Renee Purdy, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Los Angeles Region on April 16, 2020.

Renee Purdy  
Executive Officer

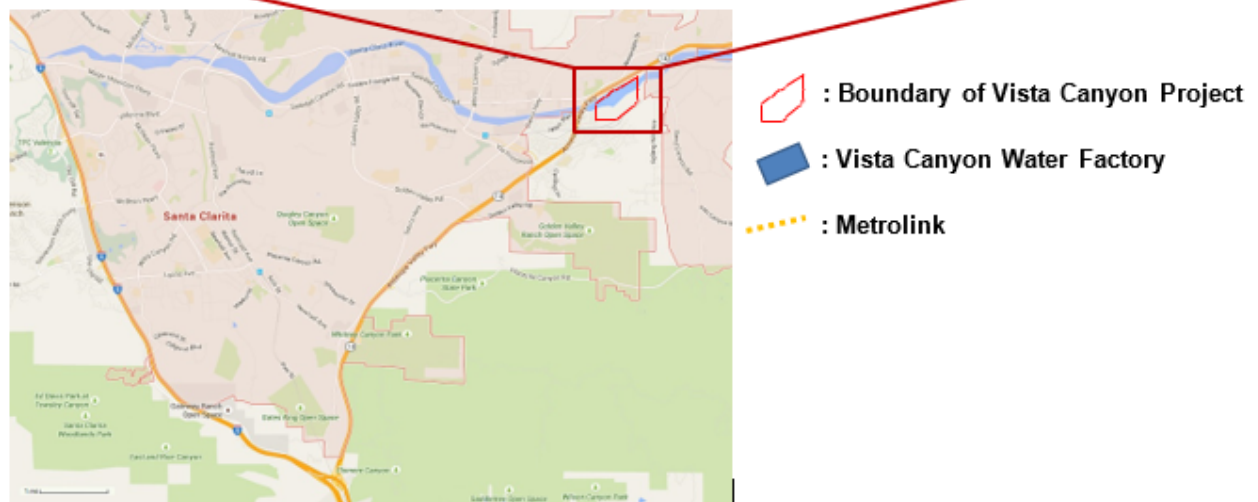


Figure 1. Vicinity of Vista Canyon Project and Vista Canyon Water Factory



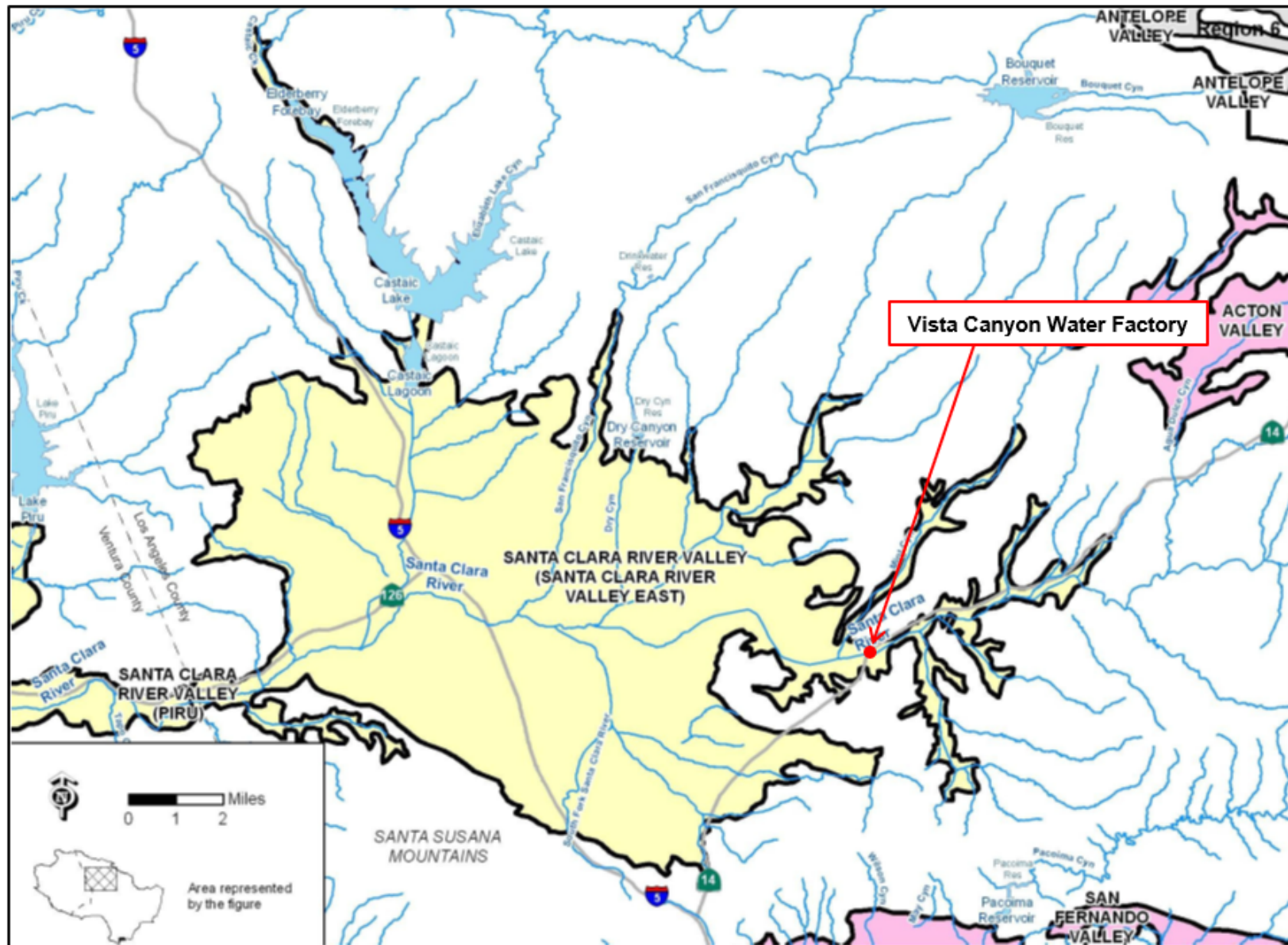


Figure 2. Santa Clara River Valley East Groundwater Basin

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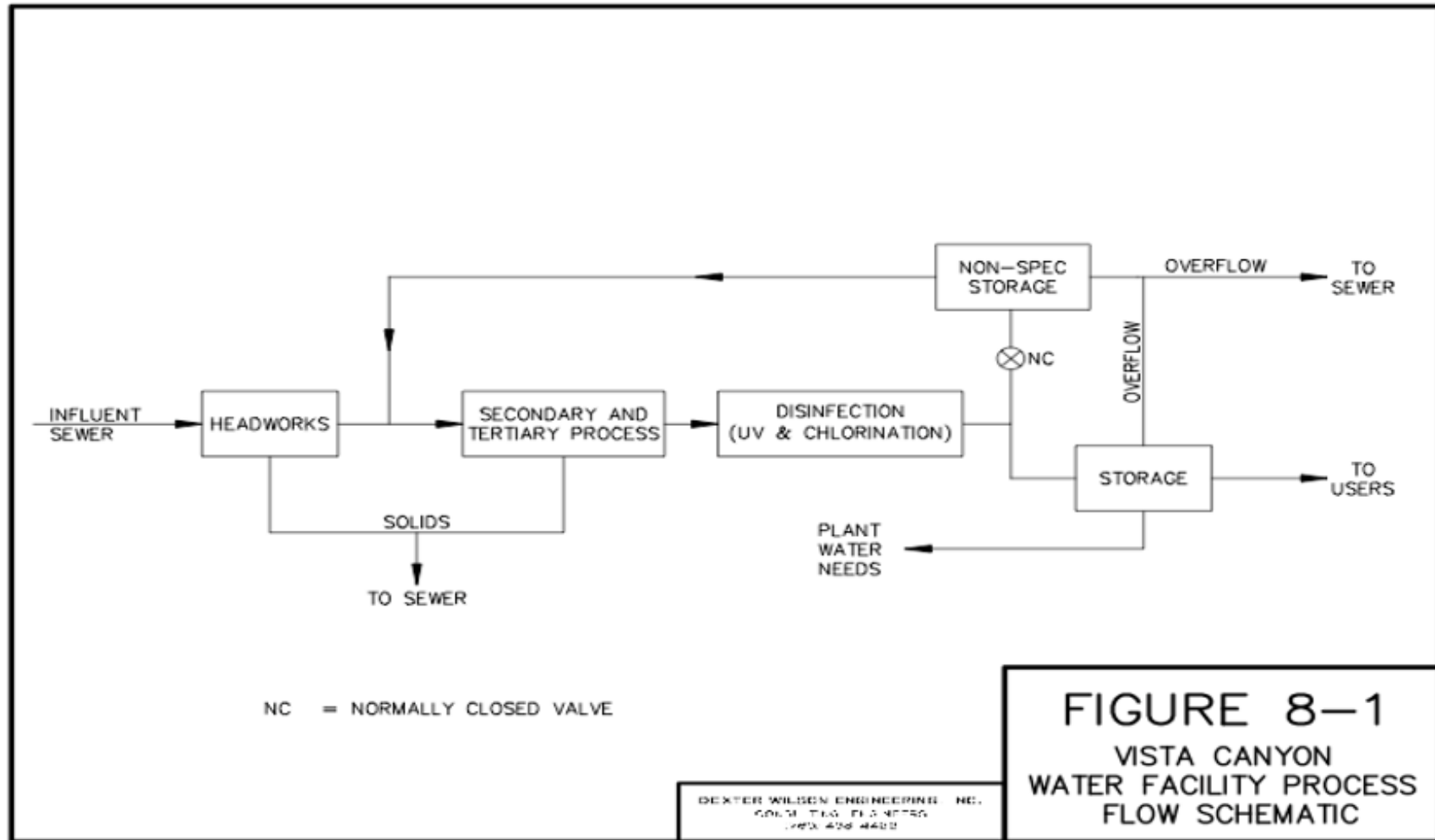


Figure 3. Process Flow Schematic of Vista Canyon Water Factory





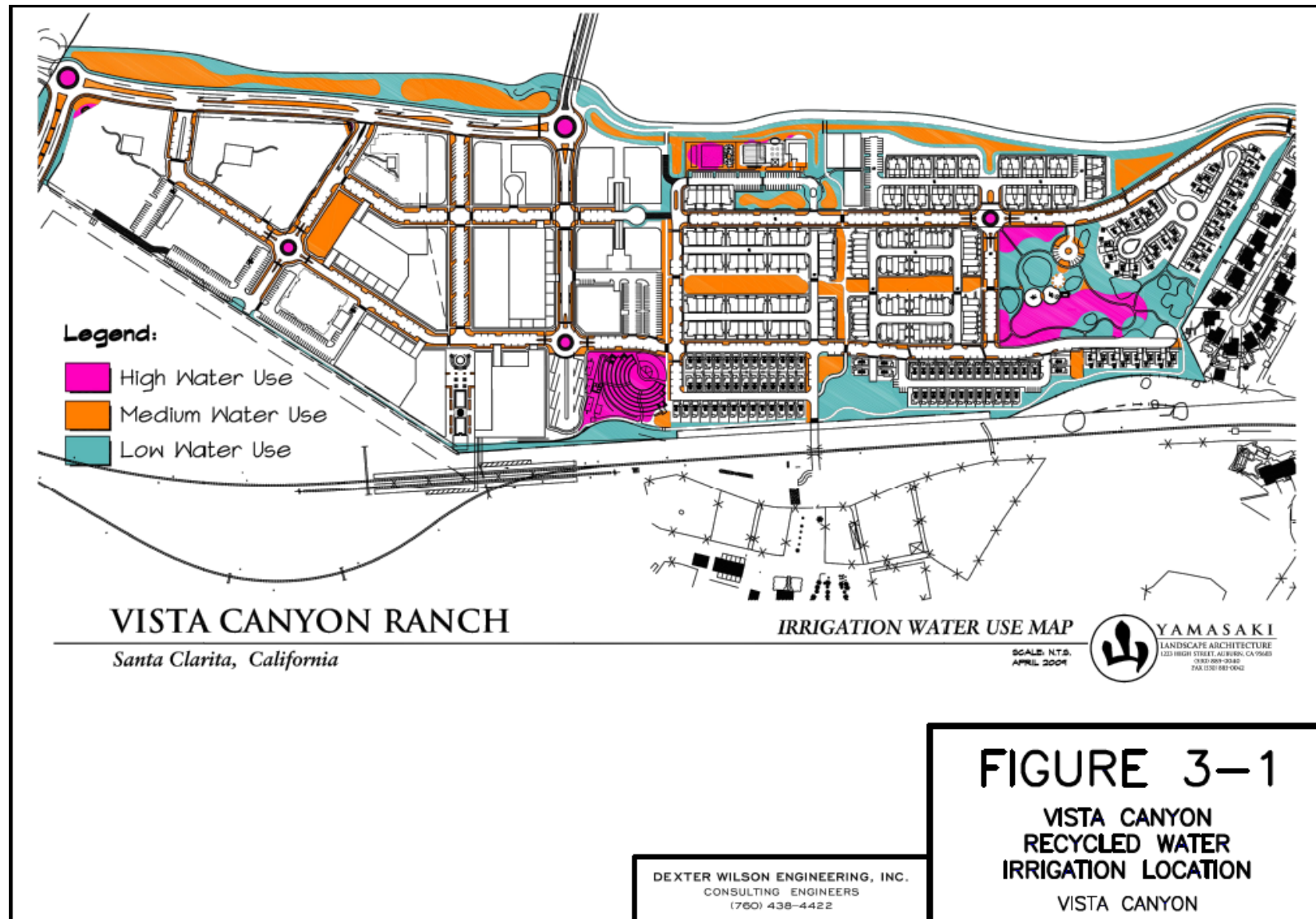


Figure 5. Locations of Potential Recycled Water Uses



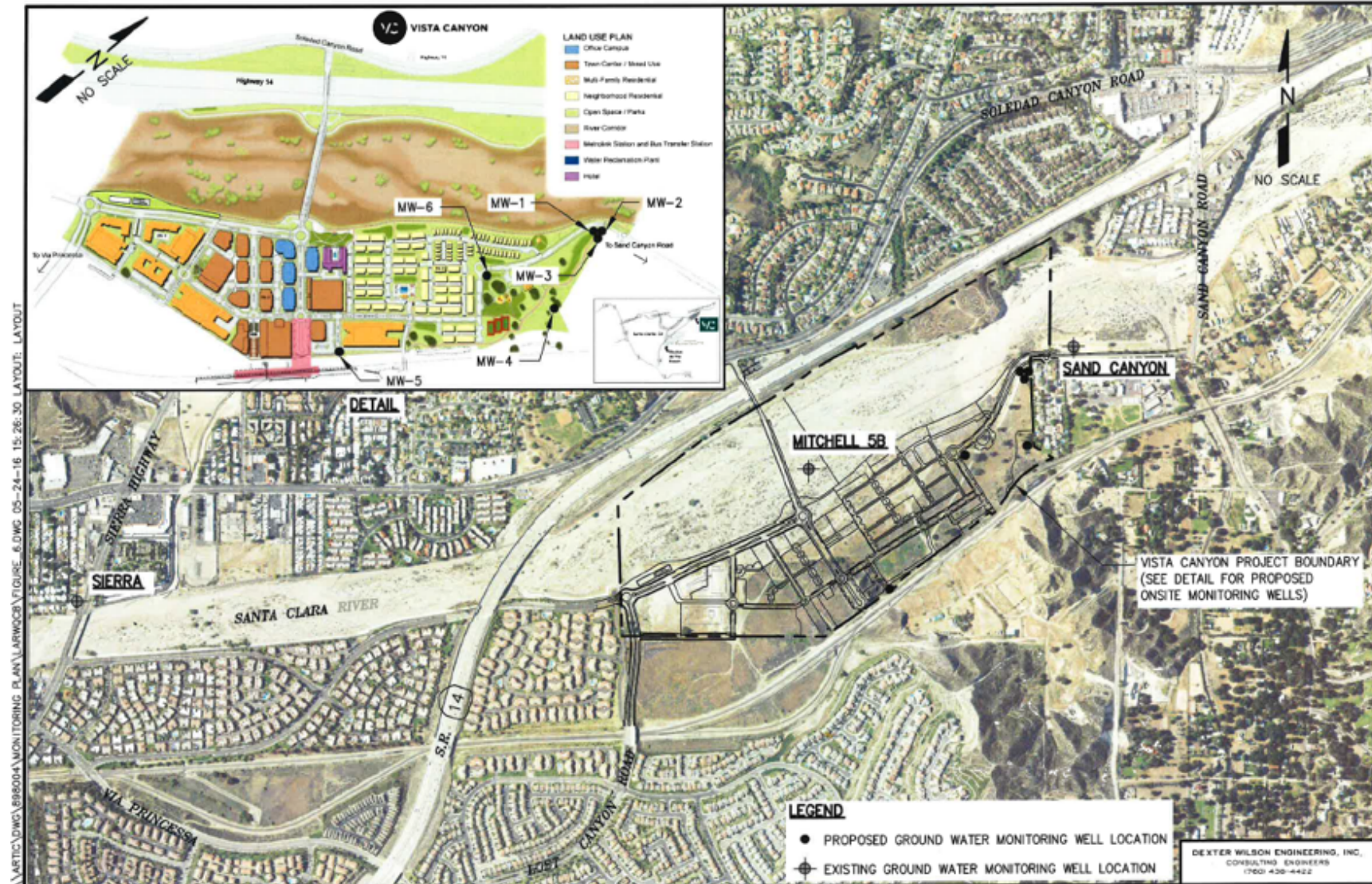


Figure 6. Locations of Groundwater Monitoring Wells for Vista Canyon Water Factory Project



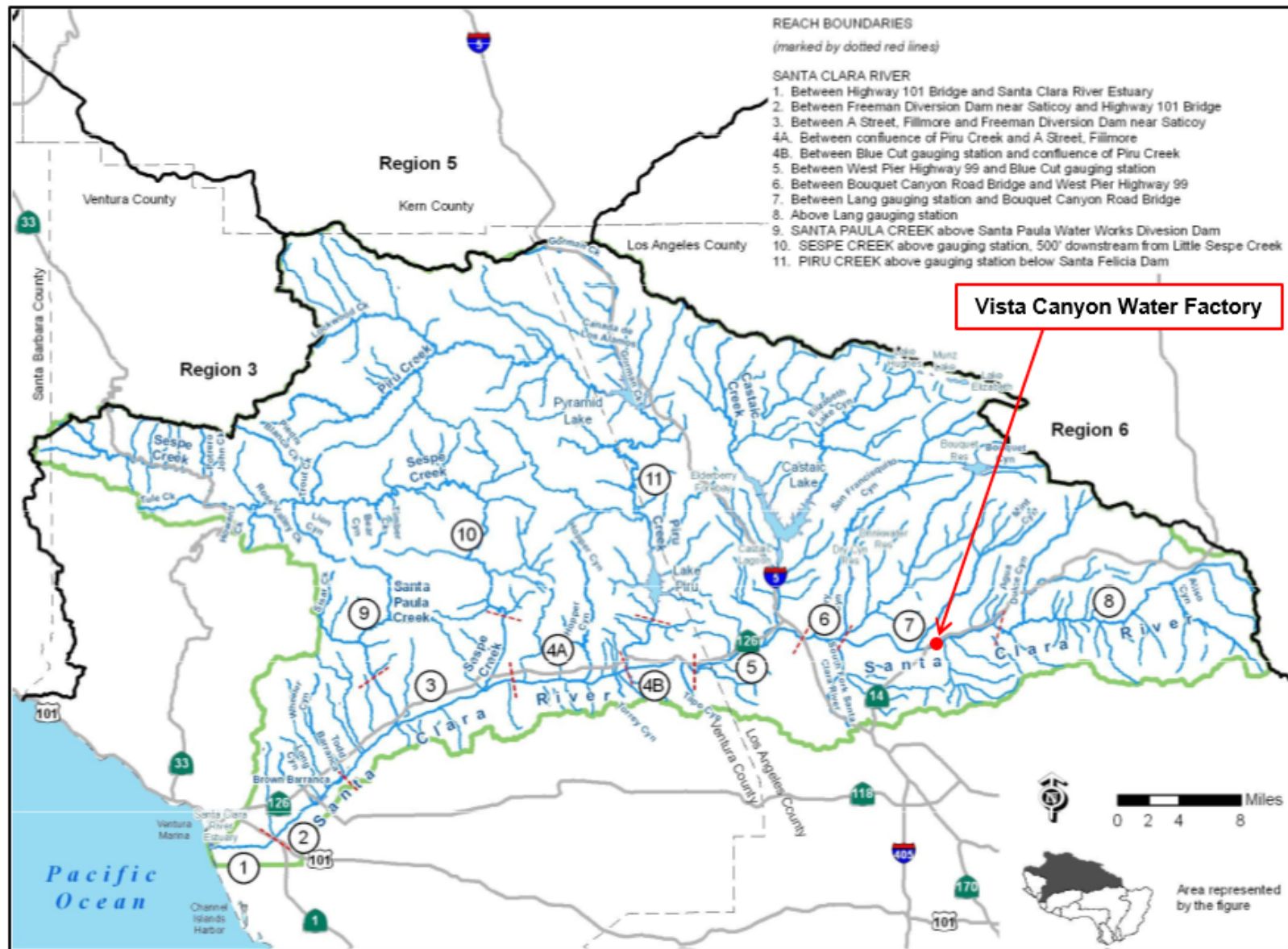


Figure 7. Santa Clara River Watershed Surface Reaches



Figure 8. Exhibit of “Recycled Water – Do Not Drink” Signage